

```
In [1]: import pandas as pd

# Load your CSV
df = pd.read_csv("VAE_file_with_uri_validation.csv")

# Invert TRUE/FALSE values
df = df.applymap(lambda x: False if str(x).upper() == "TRUE" else True if str(x).upper() == "FALSE" else x)

# Save the modified CSV
df.to_csv("inverted_VAE_file.csv", index=False)
```

```
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_7986/2356790298.py:7: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.map instead.
df = df.applymap(lambda x: False if str(x).upper() == "TRUE" else True if str(x).upper() == "FALSE" else x)
```

```
In [4]: import os
import pandas as pd
import matplotlib.pyplot as plt

# Load your CSV
df = pd.read_csv("inverted_GAN_uri_valid.csv")

# Ensure TRUE/FALSE are treated as boolean
df = df.applymap(lambda x: str(x).strip().upper() == 'TRUE')

# Folder to save images
output_folder = "GAN_donut_charts"
os.makedirs(output_folder, exist_ok=True)

# Plot and save donut charts for each column
for column in df.columns:
    counts = df[column].value_counts()
    labels = ['Valid URI', 'Literal Value']
    sizes = [counts.get(True, 0), counts.get(False, 0)]

    # Donut chart
    fig, ax = plt.subplots()
    wedges, texts, autotexts = ax.pie(
        sizes,
        labels=labels,
        autopct='%1.1f%%',
        startangle=90,
        wedgeprops={'width': 0.2}
    )
    ax.set(aspect="equal", title=f"Distribution in '{column}'")

# Save to file
filename = f"{column.replace(' ', '_')}.png"
filepath = os.path.join(output_folder, filename)
plt.savefig(filepath, bbox_inches='tight')
plt.close(fig) # Close to avoid displaying in interactive environments
```

```
print(f"Saved all donut charts to the folder: {output_folder}")
```

```
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_7986/4014127496.p  
y:9: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.ma  
p instead.
```

```
df = df.applymap(lambda x: str(x).strip().upper() == 'TRUE')
```

```
Saved all donut charts to the folder: GAN_donut_charts
```

```
In [3]: import pandas as pd
import requests
from bs4 import BeautifulSoup
import csv

# Function to fetch content from a URL
def fetch_content(url):
    try:
        response = requests.get(url, timeout=5)
        soup = BeautifulSoup(response.content, 'html.parser')
        return soup.get_text(separator='\n')
    except Exception as e:
        return f"Error: {e}"

# Function to process a single cell
def replace_uri(cell):
    if isinstance(cell, str) and cell.startswith(('http://', 'https://')):
        return fetch_content(cell)
    return cell # leave unchanged

# Try to detect the delimiter and read the CSV
try:
    with open('10_rows.csv', 'r') as file:
        dialect = csv.Sniffer().sniff(file.read(1024))
        file.seek(0) # Reset file pointer after sniffing
        df = pd.read_csv(file, delimiter=dialect.delimiter)
except Exception as e:
    print(f"Failed to read CSV: {e}")
    print("Dumping a few lines to debug:")
    with open('10_rows.csv', 'r') as f:
        lines = f.readlines()
        for i, line in enumerate(lines[:10]):
            print(f"Line {i+1}: {line.strip()}")
        raise SystemExit("Terminating due to CSV parsing error.")

# Apply the URI replacement to the entire DataFrame
df = df.applymap(replace_uri)

# Save the updated file
df.to_csv('updated_GAN_file.csv', index=False)
print("File updated and saved as 'updated_GAN_file.csv'.")
```

Failed to read CSV: Error tokenizing data. C error: Expected 4 fields in line 5, saw 5

Dumping a few lines to debug:

Line 1: Protein URI;Equivalent URI;owl:annotatedProperty;owl:annotatedSource;rdfs:comment;oboInOwl:hasBroadSynonym;oboInOwl:hasDbXref;oboInOwl:hasExactSynonym;oboInOwl:hasNarrowSynonym;oboInOwl:hasRelatedSynonym;rdfs:label;owl:onProperty;owl:someValuesFrom;rdfs:subClassOf;rdf:type;Equivalent URI_uri_valid;owl:annotatedProperty_uri_valid;owl:annotatedSource_uri_valid;rdfs:comment_uri_valid;oboInOwl:hasBroadSynonym_uri_valid;oboInOwl:hasDbXref_uri_valid;oboInOwl:hasExactSynonym_uri_valid;oboInOwl:hasNarrowSynonym_uri_valid;oboInOwl:hasRelatedSynonym_uri_valid;rdfs:label_uri_valid;owl:onProperty_uri_valid;owl:someValuesFrom_uri_valid;rdfs:subClassOf_uri_valid;rdf:type_uri_valid

Line 2: http://purl.obolibrary.org/obo/pr#Q8BP67_3; http://purl.obolibrary.org/obo/PR_Q8BP67; hPBRM1/iso:h4; http://purl.obolibrary.org/obo/PR_P39361; At2g17036; A protein coding gene acrA in Dictyostelium discoideum.; <http://www.w3.org/2002/07/owl#Axiom>; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; "http://purl.obolibrary.org/obo/PR_000022733; http://purl.obolibrary.org/obo/PR_Q9NVE5-2; PR:Q23490"; na2591bfefa654fb8a4039152cf663b4eb2431; <http://www.w3.org/2002/07/owl#Restriction>; http://purl.obolibrary.org/obo/IA0_000115; gene; A Rab GTPase-activating protein 1-like that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:Q5R372-8 or a 1:1 ortholog thereof.; "http://purl.obolibrary.org/obo/pr#Q8BP67; brevin (horse)"; FALSE; TRUE; FALSE; TRUE; TRUE; FALSE; FALSE; FALSE; TRUE; FALSE; FALSE; TRUE; TRUE; FALSE

Line 3: http://purl.obolibrary.org/obo/pr#Q9BRT8-4_5; http://purl.obolibrary.org/obo/PR_Q9BRT8-4; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; Reactome:R-HSA-6797641; "http://purl.obolibrary.org/obo/RO_0002160; gene"; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; gene; mCLP1; "http://purl.obolibrary.org/obo/PR_P0ACL5; system B(0) neutral amino acid transporter AT1 (human)"; gene; http://purl.obolibrary.org/obo/PR_000057833; "At2g17036; PR:DNx"; "N6-methyl-K395-CS; UniProtKB:Q9FFN0; gene"; UniProtKB:Q13619; "http://purl.obolibrary.org/obo/pr#Gene-based; <http://purl.obolibrary.org/obo/pr#Q9BRT8-4>"; FALSE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; TRUE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; FALSE

Line 4: http://purl.obolibrary.org/obo/pr#P31362_5; http://purl.obolibrary.org/obo/PR_P31362; <http://www.pombase.org/spombe/result/SPCC16C4.17>; Lirf; "http://www.w3.org/2002/07/owl#Axiom; ndbe7d537bd484c5da435bae1561d1cbab3857"; http://purl.obolibrary.org/obo/PR_Q6XPS3-2; "ASMT; n2bcccf89d1394cd0a28ac430d42e7938b1348"; mCLP1; http://purl.obolibrary.org/obo/PR_G3MU46-2; "http://www.w3.org/2002/07/owl#Axiom; nc3fe8eaf47d44655ace51272cb29a601b2973"; <http://zfin.org/action/marker/view/ZDB-GENE-041210-4>; "http://purl.obolibrary.org/obo/IA0_0000115; OMA:P63058"; n5717243d8f614fb483c2e84c29d52d39b2303; nff9f7ba0e0274dd396426bbe73406159b2151; "http://purl.obolibrary.org/obo/PR_000085071; <http://purl.obolibrary.org/obo/pr#P31362>"; FALSE; FALSE; TRUE; FALSE; FALSE; TRUE; TRUE; FALSE; FALSE; FALSE; FALSE; TRUE; TRUE; FALSE

Line 5: http://purl.obolibrary.org/obo/pr#Q0V869-1_2; http://purl.obolibrary.org/obo/PR_Q0V869-1; PR:DNx; <https://bar.utoronto.ca/thalemine/portal.do?externalids=AT5G49740;n97fa82eb252747c487600ada9ed0f9bab4154>; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; ne12566163ccc45b3833f59b063a82d9db838; n0c1c89fe034a4c3dbc83432e2671820ab2254; gene; n2deaf7d8b8f54dbbabd317f2c29f7865b2281; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; gen

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e;EcoGene:EG13561;UniProtKB:Q9Z0R6;"http://purl.obolibrary.org/obo/pr#Q0V869-1; http://www.w3.org/2002/07/owl#Restriction";TRUE;TRUE;FALSE;TRUE;FALSE;TRUE;TRUE;TRUE;TRUE;FALSE;TRUE;TRUE;TRUE;FALSE
Line 6: http://purl.obolibrary.org/obo/pr#Q63449-2_5;http://purl.obolibrary.org/obo/PR_Q63449-2;http://purl.obolibrary.org/obo/NCBITaxon_10116;Kif20a;n834d695861f34315918db9bc48268214b2037;"http://purl.obolibrary.org/obo/RO_0002160; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; UniProtKB:Q7ZUP1";PMID:8687384;A proliferation marker protein Ki-67 (human) that has been deaminated on the residue at the position equivalent to Gln-2020 of the amino acid sequence represented by UniProtKB:P46013-1. Example: UniProtKB:P46013-1, Gln-2020, MOD:00685.;"http://purl.obolibrary.org/obo/PR_P12367; PMID:6190803";http://www.w3.org/2002/07/owl#Axiom;na75d412feb784e458e11f5fa467ce2f3b2214;http://purl.obolibrary.org/obo/pr#Gene-based;"An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; Rab11-FIP3 (human); n0a15870cc07c46dabaa91886e3ded81bb1133";n85e2bc4e9af34d62a110d287c3dbcb29b967;"http://purl.obolibrary.org/obo/pr#Q63449-2; http://www.w3.org/2002/07/owl#Axiom";FALSE;FALSE;TRUE;TRUE;FALSE;TRUE;TRUE;FALSE;FALSE;TRUE;FALSE;TRUE;TRUE;FALSE
Line 7: http://purl.obolibrary.org/obo/pr#000031183_1;http://purl.obolibrary.org/obo/PR_000031183;"A protein coding gene nhr-66 in worm.; n5d9cb92f3d1c4eae9efa0830e4b81444b2893";http://purl.obolibrary.org/obo/PR_Q2G0B4;"http://purl.obolibrary.org/obo/pr#PR0-short-label; SUN domain-containing protein 4 (Arabidopsis thaliana)";http://www.geneontology.org/formats/oboInOwl#hasExactSynonym;nucleoside triphosphate phosphohydrolase;http://purl.obolibrary.org/obo/PR_Q84MD2;"http://www.w3.org/2002/07/owl#Restriction; na549d0dadbb4033828bb52c1c741f41b3032";http://www.w3.org/2002/07/owl#Restriction;An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.;http://purl.obolibrary.org/obo/pr#PR0-short-label;nf4b2a3a2b8074e59a5e4f5a345615854b1003;http://www.w3.org/2002/07/owl#Axiom;"http://proconsortium.org/cgi-bin/entry_pro?id=PR_000031183; http://purl.obolibrary.org/obo/pr#Gene-based";TRUE;TRUE;FALSE;FALSE;FALSE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;FALSE;TRUE;FALSE;FALSE
Line 8: http://purl.obolibrary.org/obo/pr#F1Q575_3;http://purl.obolibrary.org/obo/PR_F1Q575;mPGE synthase-2 (human);http://purl.obolibrary.org/obo/PR_Q5XKR9-2;http://purl.obolibrary.org/obo/NCBITaxon_9606;http://purl.obolibrary.org/obo/PR_Q60653;PR0:DNx;"http://www.ncbi.nlm.nih.gov/gene/1164905; FATP-2; SUN domain-containing protein 4 (Arabidopsis thaliana)";http://www.w3.org/2002/07/owl#Class;http://purl.obolibrary.org/obo/PR_000057376;http://purl.obolibrary.org/obo/PR_000070511;http://purl.obolibrary.org/obo/PR_P0C5K4;"M1 (Influenza A virus (strain A/Udorn/307/1972 H3N2)); PR0:DNx";A protein that is a translation product of the human OSGEPL1 gene or a 1:1 ortholog thereof.;"http://purl.obolibrary.org/obo/pr#F1Q575; http://www.w3.org/2002/07/owl#Class";FALSE;TRUE;FALSE;FALSE;FALSE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;TRUE;FALSE
Line 9: http://purl.obolibrary.org/obo/pr#Q9SK50_5;http://purl.obolibrary.org/obo/PR_Q9SK50;http://www.geneontology.org/formats/oboInOwl#hasExactSynonym;"Category=gene.; UniProtKB:P38070";https://bar.utoronto.ca/thalemine/porta1.do?externalids=AT4G32470;"http://purl.obolibrary.org/obo/RO_0002160; http://www.w3.org/1999/02/22-rdf-syntax-ns#nil";"http://purl.obolibrary.org/ob
```

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o/NCBITaxon_10090; http://purl.obolibrary.org/obo/PR_P70658-1; M1 (Influenza
A virus (strain A/Udorn/307/1972 H3N2));"mCLP1;"http://purl.obolibrary.org/o
bo/PR_Q6AYZ1; nf002a7abb4e346c0bee6795be1f50998b2242; nf002a7abb4e346c0bee67
95be1f50998b3678";"http://purl.obolibrary.org/obo/PR_000065167; CAALFM_C1123
60CA";http://purl.obolibrary.org/obo/PR_Q8K0D0;gene;"Aqr (mouse); nf434fd27b
84b4345befbd5526f317f54b1061";"A protein coding gene ipo8 in zebrafish.; n4c
641700d6bb46049e644d5ff510cf6cb3402";"http://purl.obolibrary.org/obo/pr#Q9SK
50; http://www.w3.org/2002/07/owl#Class";TRUE;FALSE;TRUE;FALSE;FALSE;FALSE;T
RUE;FALSE;FALSE;FALSE;TRUE;TRUE;TRUE;FALSE
Line 10: http://purl.obolibrary.org/obo/pr#000029949_3;http://purl.obolibrar
y.org/obo/PR_000029949;http://purl.obolibrary.org/obo/PR_Q60653;nd7f01bd2140
640c5b620ac65dcb0d808b1459;n6b50d633409242829b2df002a77ad435b569;http://pur
l.obolibrary.org/obo/PR_000008220;"http://purl.obolibrary.org/obo/PR_D3ZUG8;
http://purl.obolibrary.org/obo/PR_Q84LK0";M1 (Influenza A virus (strain A/Ud
orn/307/1972 H3N2));"NVP-2 (rat); Rab11-FIP3 (human)";http://www.genenames.o
rg/cgi-bin/gene_symbol_report?hgnc_id=9863;http://www.geneontology.org/forma
ts/oboInOwl#hasRelatedSynonym;http://purl.obolibrary.org/obo/PR_D4AE48;htt
p://purl.obolibrary.org/obo/PR_P07283;SUN domain-containing protein 4 (Arabi
dopsis thaliana);"https://proconsortium.org/cgi-bin/entry_pro?id=PR_00002994
9; http://purl.obolibrary.org/obo/pr#PR0-short-label";FALSE;FALSE;TRUE;TRUE;
FALSE;FALSE;TRUE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;FALSE

```

An exception has occurred, use %tb to see the full traceback.

SystemExit: Terminating due to CSV parsing error.

```

/opt/anaconda3/lib/python3.12/site-packages/IPython/core/interactiveshell.p
y:3585: UserWarning: To exit: use 'exit', 'quit', or Ctrl-D.
warn("To exit: use 'exit', 'quit', or Ctrl-D.", stacklevel=1)

```

```

In [4]: with open("GAN_sampled_200_rows_copy.csv") as f:
        for i, line in enumerate(f):
            print(f"Line {i+1} has {len(line.split('\t'))} columns")

```

Line 1 has 1 columns
Line 2 has 1 columns
Line 3 has 1 columns
Line 4 has 1 columns
Line 5 has 1 columns
Line 6 has 1 columns
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```
In [5]: import pandas as pd

df = pd.read_csv('GAN_sampled_200_rows copy.csv', sep=None, engine='python')
print(df.head())
```

```

                                Protein URI \
0   http://purl.obolibrary.org/obo/pr#Q8BP67_3
1   http://purl.obolibrary.org/obo/pr#Q9BRT8-4_5
2   http://purl.obolibrary.org/obo/pr#P31362_5
3   http://purl.obolibrary.org/obo/pr#Q0V869-1_2
4   http://purl.obolibrary.org/obo/pr#Q63449-2_5

                                Equivalent URI \
0   http://purl.obolibrary.org/obo/PR_Q8BP67
1   http://purl.obolibrary.org/obo/PR_Q9BRT8-4
2   http://purl.obolibrary.org/obo/PR_P31362
3   http://purl.obolibrary.org/obo/PR_Q0V869-1
4   http://purl.obolibrary.org/obo/PR_Q63449-2

                                owl:annotatedProperty \
0                                     hPBRM1/iso:h4
1   http://www.geneontology.org/formats/oboInOwl#h...
2   http://www.pombase.org/spombe/result/SPCC16C4.17
3                                     PRO:DNx
4   http://purl.obolibrary.org/obo/NCBITaxon_10116

                                owl:annotatedSource \
0   http://purl.obolibrary.org/obo/PR_P39361
1   Reactome:R-HSA-6797641
2   Lirf
3   https://bar.utoronto.ca/thalemine/portal.do?ex...
4   Kif20a

                                rdfs:comment \
0   At2g17036
1   http://purl.obolibrary.org/obo/R0_0002160; gene
2   http://www.w3.org/2002/07/owl#Axiom; ndbe7d537...
3   n97fa82eb252747c487600ada9ed0f9bab4154
4   n834d695861f34315918db9bc48268214b2037

                                oboInOwl:hasBroadSynonym \
0   A protein coding gene acrA in Dictyostelium di...
1   An interleukin-1 receptor type 1 that is a tra...
2   http://purl.obolibrary.org/obo/PR_Q6XPS3-2
3   http://www.geneontology.org/formats/oboInOwl#h...
4   http://purl.obolibrary.org/obo/R0_0002160; An ...

                                oboInOwl:hasDbXref \
0   http://www.w3.org/2002/07/owl#Axiom
1   gene
2   ASMT; n2bccc89d1394cd0a28ac430d42e7938b1348
3   ne12566163ccc45b3833f59b063a82d9db838
4   PMID:8687384

                                oboInOwl:hasExactSynonym \
0   http://www.geneontology.org/formats/oboInOwl#h...
1   mCLP1
2   mCLP1
3   n0c1c89fe034a4c3dbc83432e2671820ab2254
4   A proliferation marker protein Ki-67 (human) t...

```

	oboInOwl:hasNarrowSynonym \
0	http://purl.obolibrary.org/obo/PR_000022733; h...
1	http://purl.obolibrary.org/obo/PR_P0ACL5; syst...
2	http://purl.obolibrary.org/obo/PR_G3MU46-2
3	gene
4	http://purl.obolibrary.org/obo/PR_P12367; PMID...

	oboInOwl:hasRelatedSynonym ... \
0	na2591bfefa654fb8a4039152cf663b4eb2431 ...
1	gene ...
2	http://www.w3.org/2002/07/owl#Axiom; nc3fe8eaf... ...
3	n2deaf7d8b8f54dbbabb317f2c29f7865b2281 ...
4	http://www.w3.org/2002/07/owl#Axiom ...

	oboInOwl:hasBroadSynonym_uri_valid	oboInOwl:hasDbXref_uri_valid \
0	True	False
1	True	True
2	False	True
3	False	True
4	False	True

	oboInOwl:hasExactSynonym_uri_valid	oboInOwl:hasNarrowSynonym_uri_valid \
0	False	False
1	True	False
2	True	False
3	True	True
4	True	False

	oboInOwl:hasRelatedSynonym_uri_valid	rdfs:label_uri_valid \
0	True	False
1	True	False
2	False	False
3	True	False
4	False	True

	owl:onProperty_uri_valid	owl:someValuesFrom_uri_valid \
0	False	True
1	True	True
2	False	True
3	True	True
4	False	True

	rdfs:subClassOf_uri_valid	rdf:type_uri_valid
0	True	False
1	True	False
2	True	False
3	True	False
4	True	False

[5 rows x 29 columns]

```
In [6]: import pandas as pd
import requests
from bs4 import BeautifulSoup
import csv
```

```

# Function to fetch content from a URL
def fetch_content(url):
    try:
        response = requests.get(url, timeout=5)
        soup = BeautifulSoup(response.content, 'html.parser')
        return soup.get_text(separator='\n')
    except Exception as e:
        return f"Error: {e}"

# Function to process a single cell
def replace_uri(cell):
    if isinstance(cell, str) and cell.startswith(('http://', 'https://')):
        return fetch_content(cell)
    return cell # leave unchanged

# Try to detect the delimiter and read the CSV
try:
    with open('10_rows.csv', 'r') as file:
        dialect = csv.Sniffer().sniff(file.read(1024))
        file.seek(0) # Reset file pointer after sniffing
        df = pd.read_csv(file, delimiter=dialect.delimiter)
except Exception as e:
    print(f"Failed to read CSV: {e}")
    print("Dumping a few lines to debug:")
    with open('10_rows.csv', 'r') as f:
        lines = f.readlines()
        for i, line in enumerate(lines[:10]):
            print(f"Line {i+1}: {line.strip()}")
        raise SystemExit("Terminating due to CSV parsing error.")

# Apply the URI replacement to the entire DataFrame
df = df.applymap(replace_uri)

# Save the updated file
df.to_csv('updated_GAN_file.csv', index=False)
print("File updated and saved as 'updated_GAN_file.csv'.")

```

Failed to read CSV: Error tokenizing data. C error: Expected 4 fields in line 5, saw 5

Dumping a few lines to debug:

Line 1: Protein URI;Equivalent URI;owl:annotatedProperty;owl:annotatedSource;rdfs:comment;oboInOwl:hasBroadSynonym;oboInOwl:hasDbXref;oboInOwl:hasExactSynonym;oboInOwl:hasNarrowSynonym;oboInOwl:hasRelatedSynonym;rdfs:label;owl:onProperty;owl:someValuesFrom;rdfs:subClassOf;rdf:type;Equivalent URI_uri_valid;owl:annotatedProperty_uri_valid;owl:annotatedSource_uri_valid;rdfs:comment_uri_valid;oboInOwl:hasBroadSynonym_uri_valid;oboInOwl:hasDbXref_uri_valid;oboInOwl:hasExactSynonym_uri_valid;oboInOwl:hasNarrowSynonym_uri_valid;oboInOwl:hasRelatedSynonym_uri_valid;rdfs:label_uri_valid;owl:onProperty_uri_valid;owl:someValuesFrom_uri_valid;rdfs:subClassOf_uri_valid;rdf:type_uri_valid

Line 2: http://purl.obolibrary.org/obo/pr#Q8BP67_3; http://purl.obolibrary.org/obo/PR_Q8BP67; hPBRM1/iso:h4; http://purl.obolibrary.org/obo/PR_P39361; At2g17036; A protein coding gene acrA in Dictyostelium discoideum.; <http://www.w3.org/2002/07/owl#Axiom>; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; "http://purl.obolibrary.org/obo/PR_000022733; http://purl.obolibrary.org/obo/PR_Q9NVE5-2; PR:Q23490"; na2591bfefa654fb8a4039152cf663b4eb2431; <http://www.w3.org/2002/07/owl#Restriction>; http://purl.obolibrary.org/obo/IA0_000115; gene; A Rab GTPase-activating protein 1-like that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:Q5R372-8 or a 1:1 ortholog thereof.; "http://purl.obolibrary.org/obo/pr#Q8BP67; brevin (horse)"; FALSE; TRUE; FALSE; TRUE; TRUE; FALSE; FALSE; FALSE; TRUE; FALSE; FALSE; TRUE; TRUE; FALSE

Line 3: http://purl.obolibrary.org/obo/pr#Q9BRT8-4_5; http://purl.obolibrary.org/obo/PR_Q9BRT8-4; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; Reactome:R-HSA-6797641; "http://purl.obolibrary.org/obo/RO_0002160; gene"; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; gene; mCLP1; "http://purl.obolibrary.org/obo/PR_P0ACL5; system B(0) neutral amino acid transporter AT1 (human)"; gene; http://purl.obolibrary.org/obo/PR_000057833; "At2g17036; PR:DNx"; "N6-methyl-K395-CS; UniProtKB:Q9FFN0; gene"; UniProtKB:Q13619; "http://purl.obolibrary.org/obo/pr#Gene-based; <http://purl.obolibrary.org/obo/pr#Q9BRT8-4>"; FALSE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; TRUE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; FALSE

Line 4: http://purl.obolibrary.org/obo/pr#P31362_5; http://purl.obolibrary.org/obo/PR_P31362; <http://www.pombase.org/spombe/result/SPCC16C4.17>; Lirf; "http://www.w3.org/2002/07/owl#Axiom; ndbe7d537bd484c5da435bae1561d1cbab3857"; http://purl.obolibrary.org/obo/PR_Q6XPS3-2; "ASMT; n2bcccf89d1394cd0a28ac430d42e7938b1348"; mCLP1; http://purl.obolibrary.org/obo/PR_G3MU46-2; "http://www.w3.org/2002/07/owl#Axiom; nc3fe8eaf47d44655ace51272cb29a601b2973"; <http://zfin.org/action/marker/view/ZDB-GENE-041210-4>; "http://purl.obolibrary.org/obo/IA0_0000115; OMA:P63058"; n5717243d8f614fb483c2e84c29d52d39b2303; nff9f7ba0e0274dd396426bbe73406159b2151; "http://purl.obolibrary.org/obo/PR_000085071; <http://purl.obolibrary.org/obo/pr#P31362>"; FALSE; FALSE; TRUE; FALSE; FALSE; TRUE; TRUE; FALSE; FALSE; FALSE; FALSE; TRUE; TRUE; FALSE

Line 5: http://purl.obolibrary.org/obo/pr#Q0V869-1_2; http://purl.obolibrary.org/obo/PR_Q0V869-1; PR:DNx; <https://bar.utoronto.ca/thalemine/portal.do?externalids=AT5G49740;n97fa82eb252747c487600ada9ed0f9bab4154>; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; ne12566163ccc45b3833f59b063a82d9db838;n0c1c89fe034a4c3dbc83432e2671820ab2254; gene; n2deaf7d8b8f54dbbabd317f2c29f7865b2281; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; gen

```
e;EcoGene:EG13561;UniProtKB:Q9Z0R6;"http://purl.obolibrary.org/obo/pr#Q0V869-1; http://www.w3.org/2002/07/owl#Restriction";TRUE;TRUE;FALSE;TRUE;FALSE;TRUE;TRUE;TRUE;TRUE;FALSE;TRUE;TRUE;TRUE;FALSE
Line 6: http://purl.obolibrary.org/obo/pr#Q63449-2_5;http://purl.obolibrary.org/obo/PR_Q63449-2;http://purl.obolibrary.org/obo/NCBITaxon_10116;Kif20a;n834d695861f34315918db9bc48268214b2037;"http://purl.obolibrary.org/obo/RO_0002160; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; UniProtKB:Q7ZUP1";PMID:8687384;A proliferation marker protein Ki-67 (human) that has been deaminated on the residue at the position equivalent to Gln-2020 of the amino acid sequence represented by UniProtKB:P46013-1. Example: UniProtKB:P46013-1, Gln-2020, MOD:00685.;"http://purl.obolibrary.org/obo/PR_P12367; PMID:6190803";http://www.w3.org/2002/07/owl#Axiom;na75d412feb784e458e11f5fa467ce2f3b2214;http://purl.obolibrary.org/obo/pr#Gene-based;"An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; Rab11-FIP3 (human); n0a15870cc07c46dabaa91886e3ded81bb1133";n85e2bc4e9af34d62a110d287c3dbcb29b967;"http://purl.obolibrary.org/obo/pr#Q63449-2; http://www.w3.org/2002/07/owl#Axiom";FALSE;FALSE;TRUE;TRUE;FALSE;TRUE;TRUE;FALSE;FALSE;TRUE;FALSE;TRUE;TRUE;FALSE
Line 7: http://purl.obolibrary.org/obo/pr#000031183_1;http://purl.obolibrary.org/obo/PR_000031183;"A protein coding gene nhr-66 in worm.; n5d9cb92f3d1c4eae9efa0830e4b81444b2893";http://purl.obolibrary.org/obo/PR_Q2G0B4;"http://purl.obolibrary.org/obo/pr#PR0-short-label; SUN domain-containing protein 4 (Arabidopsis thaliana)";http://www.geneontology.org/formats/oboInOwl#hasExactSynonym;nucleoside triphosphate phosphohydrolase;http://purl.obolibrary.org/obo/PR_Q84MD2;"http://www.w3.org/2002/07/owl#Restriction; na549d0dadbb4033828bb52c1c741f41b3032";http://www.w3.org/2002/07/owl#Restriction;An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.;http://purl.obolibrary.org/obo/pr#PR0-short-label;nf4b2a3a2b8074e59a5e4f5a345615854b1003;http://www.w3.org/2002/07/owl#Axiom;"http://proconsortium.org/cgi-bin/entry_pro?id=PR_000031183; http://purl.obolibrary.org/obo/pr#Gene-based";TRUE;TRUE;FALSE;FALSE;FALSE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;FALSE;TRUE;FALSE;FALSE
Line 8: http://purl.obolibrary.org/obo/pr#F1Q575_3;http://purl.obolibrary.org/obo/PR_F1Q575;mPGE synthase-2 (human);http://purl.obolibrary.org/obo/PR_Q5XKR9-2;http://purl.obolibrary.org/obo/NCBITaxon_9606;http://purl.obolibrary.org/obo/PR_Q60653;PR0:DNx;"http://www.ncbi.nlm.nih.gov/gene/1164905; FATP-2; SUN domain-containing protein 4 (Arabidopsis thaliana)";http://www.w3.org/2002/07/owl#Class;http://purl.obolibrary.org/obo/PR_000057376;http://purl.obolibrary.org/obo/PR_000070511;http://purl.obolibrary.org/obo/PR_P0C5K4;"M1 (Influenza A virus (strain A/Udorn/307/1972 H3N2)); PR0:DNx";A protein that is a translation product of the human OSGEPL1 gene or a 1:1 ortholog thereof.;"http://purl.obolibrary.org/obo/pr#F1Q575; http://www.w3.org/2002/07/owl#Class";FALSE;TRUE;FALSE;FALSE;FALSE;TRUE;FALSE;FALSE;FALSE;FALSE;FALSE;TRUE;TRUE;FALSE
Line 9: http://purl.obolibrary.org/obo/pr#Q9SK50_5;http://purl.obolibrary.org/obo/PR_Q9SK50;http://www.geneontology.org/formats/oboInOwl#hasExactSynonym;"Category=gene.; UniProtKB:P38070";https://bar.utoronto.ca/thalemine/porta1.do?externalids=AT4G32470;"http://purl.obolibrary.org/obo/RO_0002160; http://www.w3.org/1999/02/22-rdf-syntax-ns#nil";"http://purl.obolibrary.org/ob
```

```

o/NCBITaxon_10090; http://purl.obolibrary.org/obo/PR_P70658-1; M1 (Influenza
A virus (strain A/Udorn/307/1972 H3N2));"mCLP1;"http://purl.obolibrary.org/o
bo/PR_Q6AYZ1; nf002a7abb4e346c0bee6795be1f50998b2242; nf002a7abb4e346c0bee67
95be1f50998b3678";"http://purl.obolibrary.org/obo/PR_000065167; CAALFM_C1123
60CA";http://purl.obolibrary.org/obo/PR_Q8K0D0;gene;"Aqr (mouse); nf434fd27b
84b4345befbd5526f317f54b1061";"A protein coding gene ipo8 in zebrafish.; n4c
641700d6bb46049e644d5ff510cf6cb3402";"http://purl.obolibrary.org/obo/pr#Q9SK
50; http://www.w3.org/2002/07/owl#Class";TRUE;FALSE;TRUE;FALSE;FALSE;FALSE;T
RUE;FALSE;FALSE;FALSE;TRUE;TRUE;TRUE;FALSE
Line 10: http://purl.obolibrary.org/obo/pr#000029949_3;http://purl.obolibrar
y.org/obo/PR_000029949;http://purl.obolibrary.org/obo/PR_Q60653;nd7f01bd2140
640c5b620ac65dcb0d808b1459;n6b50d633409242829b2df002a77ad435b569;http://pur
l.obolibrary.org/obo/PR_000008220;"http://purl.obolibrary.org/obo/PR_D3ZUG8;
http://purl.obolibrary.org/obo/PR_Q84LK0";M1 (Influenza A virus (strain A/Ud
orn/307/1972 H3N2));"NVP-2 (rat); Rab11-FIP3 (human)";http://www.genenames.o
rg/cgi-bin/gene_symbol_report?hgnc_id=9863;http://www.geneontology.org/forma
ts/oboInOwl#hasRelatedSynonym;http://purl.obolibrary.org/obo/PR_D4AE48;htt
p://purl.obolibrary.org/obo/PR_P07283;SUN domain-containing protein 4 (Arabi
dopsis thaliana);"https://proconsortium.org/cgi-bin/entry_pro?id=PR_00002994
9; http://purl.obolibrary.org/obo/pr#PR0-short-label";FALSE;FALSE;TRUE;TRUE;
FALSE;FALSE;TRUE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;FALSE

```

An exception has occurred, use %tb to see the full traceback.

SystemExit: Terminating due to CSV parsing error.

```

/opt/anaconda3/lib/python3.12/site-packages/IPython/core/interactiveshell.p
y:3585: UserWarning: To exit: use 'exit', 'quit', or Ctrl-D.
warn("To exit: use 'exit', 'quit', or Ctrl-D.", stacklevel=1)

```

```

In [9]: import pandas as pd
import requests
from bs4 import BeautifulSoup
import csv

def fetch_content(url):
    try:
        response = requests.get(url, timeout=5)
        response.raise_for_status()
        soup = BeautifulSoup(response.content, 'html.parser')
        return soup.get_text(separator='\n')
    except Exception as e:
        return f"Error fetching URL: {e}"

def replace_uri(cell):
    if isinstance(cell, str) and cell.startswith(('http://', 'https://')):
        return fetch_content(cell)
    return cell

try:
    with open('10_rows.csv', 'r', encoding='utf-8') as file:
        sample = file.read(1024)
        dialect = csv.Sniffer().sniff(sample, delimiters=';,,')
        file.seek(0)
        df = pd.read_csv(file, delimiter=dialect.delimiter)
except Exception as e:
    print(f"Failed to read CSV: {e}")

```

```
with open('10_rows.csv', 'r', encoding='utf-8') as f:
    lines = f.readlines()
    for i, line in enumerate(lines[:10]):
        print(f"Line {i+1}: {line.strip()}")
    raise SystemExit("Terminating due to CSV parsing error.")

df = df.applymap(replace_uri)
df.to_csv('updated_GAN_file.csv', index=False)
print("File updated and saved as 'updated_GAN_file.csv'.")
```


Failed to read CSV: Could not determine delimiter

Line 1: Protein URI;Equivalent URI;owl:annotatedProperty;owl:annotatedSource;rdfs:comment;oboInOwl:hasBroadSynonym;oboInOwl:hasDbXref;oboInOwl:hasExactSynonym;oboInOwl:hasNarrowSynonym;oboInOwl:hasRelatedSynonym;rdfs:label;owl:onProperty;owl:someValuesFrom;rdfs:subClassOf;rdf:type;Equivalent URI_uri_valid;owl:annotatedProperty_uri_valid;owl:annotatedSource_uri_valid;rdfs:comment_uri_valid;oboInOwl:hasBroadSynonym_uri_valid;oboInOwl:hasDbXref_uri_valid;oboInOwl:hasExactSynonym_uri_valid;oboInOwl:hasNarrowSynonym_uri_valid;oboInOwl:hasRelatedSynonym_uri_valid;rdfs:label_uri_valid;owl:onProperty_uri_valid;owl:someValuesFrom_uri_valid;rdfs:subClassOf_uri_valid;rdf:type_uri_valid

Line 2: http://purl.obolibrary.org/obo/pr#Q8BP67_3; http://purl.obolibrary.org/obo/PR_Q8BP67; hPBRM1; iso:h4; http://purl.obolibrary.org/obo/PR_P39361; At2g17036; A protein coding gene acrA in Dictyostelium discoideum.; <http://www.w3.org/2002/07/owl#Axiom>; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; "http://purl.obolibrary.org/obo/PR_000022733; http://purl.obolibrary.org/obo/PR_Q9NVE5-2; PR:Q23490"; na2591bfefa654fb8a4039152cf663b4eb2431; <http://www.w3.org/2002/07/owl#Restriction>; http://purl.obolibrary.org/obo/IA0_0000115; gene; A Rab GTPase-activating protein 1-like that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:Q5R372-8 or a 1:1 ortholog thereof.; "http://purl.obolibrary.org/obo/pr#Q8BP67; brevin (horse)"; FALSE; TRUE; FALSE; TRUE; TRUE; FALSE; FALSE; FALSE; TRUE; FALSE; FALSE; TRUE; TRUE; FALSE

Line 3: http://purl.obolibrary.org/obo/pr#Q9BRT8-4_5; http://purl.obolibrary.org/obo/PR_Q9BRT8-4; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; Reactome:R-HSA-6797641; "http://purl.obolibrary.org/obo/RO_0002160; gene"; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; gene; mCLP1; "http://purl.obolibrary.org/obo/PR_P0ACL5; system B(0) neutral amino acid transporter AT1 (human)"; gene; http://purl.obolibrary.org/obo/PR_000057833; "At2g17036; PR0:DNx"; "N6-methyl-K395-CS; UniProtKB:Q9FFN0; gene"; UniProtKB:Q13619; "http://purl.obolibrary.org/obo/pr#Gene-based; <http://purl.obolibrary.org/obo/pr#Q9BRT8-4>"; FALSE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; FALSE

Line 4: http://purl.obolibrary.org/obo/pr#P31362_5; http://purl.obolibrary.org/obo/PR_P31362; <http://www.pombase.org/spombe/result/SPCC16C4.17>; Lirf; "http://www.w3.org/2002/07/owl#Axiom; ndbe7d537bd484c5da435bae1561d1cbab3857"; http://purl.obolibrary.org/obo/PR_Q6XPS3-2; "ASMT; n2bccc89d1394cd0a28ac430d42e7938b1348"; mCLP1; http://purl.obolibrary.org/obo/PR_G3MU46-2; "http://www.w3.org/2002/07/owl#Axiom; nc3fe8eaf47d44655ace51272cb29a601b2973"; <http://zfin.org/action/marker/view/ZDB-GENE-041210-4>; "http://purl.obolibrary.org/obo/IA0_0000115; OMA:P63058"; n5717243d8f614fb483c2e84c29d52d39b2303; nff9f7ba0e0274dd396426bbe73406159b2151; "http://purl.obolibrary.org/obo/PR_000085071; <http://purl.obolibrary.org/obo/pr#P31362>"; FALSE; FALSE; TRUE; FALSE; FALSE; TRUE; TRUE; FALSE; FALSE; FALSE; FALSE; TRUE; TRUE; FALSE

Line 5: http://purl.obolibrary.org/obo/pr#Q0V869-1_2; http://purl.obolibrary.org/obo/PR_Q0V869-1; PR0:DNx; <https://bar.utoronto.ca/thalemine/portal.do?externalids=AT5G49740;n97fa82eb252747c487600ada9ed0f9bab4154>; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; ne12566163ccc45b3833f59b063a82d9db838; n0c1c89fe034a4c3dbc83432e2671820ab2254; gene; n2deaf7d8b8f54dbbabd317f2c29f7865b2281; <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>; gene; EcoGene:EG13561; UniProtKB:Q9Z0R6; "http://purl.obolibrary.org/obo/pr#Q0V869-1; <http://www.w3.org/2002/07/owl#Restriction>"; TRUE; TRUE; FALSE; TRUE; FALSE; TRUE; TRUE; TRUE; TRUE; FALSE; TRUE; TRUE; TRUE; FALSE

Line 6: http://purl.obolibrary.org/obo/pr#Q63449-2_5;http://purl.obolibrary.org/obo/PR_Q63449-2;http://purl.obolibrary.org/obo/NCBITaxon_10116;Kif20a;n834d695861f34315918db9bc48268214b2037;"http://purl.obolibrary.org/obo/R0_0002160; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; UniProtKB:Q7ZUP1";PMID:8687384;A proliferation marker protein Ki-67 (human) that has been deaminated on the residue at the position equivalent to Gln-2020 of the amino acid sequence represented by UniProtKB:P46013-1. Example: UniProtKB:P46013-1, Gln-2020, MOD:00685.;"http://purl.obolibrary.org/obo/PR_P12367; PMID:6190803";<http://www.w3.org/2002/07/owl#Axiom>;na75d412feb784e458e11f5fa467ce2f3b2214;<http://purl.obolibrary.org/obo/pr#Gene-based>;"An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; Rab11-FIP3 (human); n0a15870cc07c46dabaa91886e3ded81bb1133";n85e2bc4e9af34d62a110d287c3dbcb29b967;"<http://purl.obolibrary.org/obo/pr#Q63449-2>; <http://www.w3.org/2002/07/owl#Axiom>";FALSE;FALSE;TRUE;TRUE;FALSE;TRUE;TRUE;FALSE;FALSE;TRUE;FALSE;TRUE;TRUE;FALSE

Line 7: http://purl.obolibrary.org/obo/pr#000031183_1;http://purl.obolibrary.org/obo/PR_000031183;"A protein coding gene nhr-66 in worm.; n5d9cb92f3d1c4eae9efa0830e4b81444b2893";http://purl.obolibrary.org/obo/PR_Q2G0B4;"<http://purl.obolibrary.org/obo/pr#PR0-short-label>; SUN domain-containing protein 4 (Arabidopsis thaliana)";<http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>;nucleoside triphosphate phosphohydrolase;http://purl.obolibrary.org/obo/PR_Q84MD2;"<http://www.w3.org/2002/07/owl#Restriction>; na549d0dadbb4033828bb52c1c741f41b3032";<http://www.w3.org/2002/07/owl#Restriction>;An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.;<http://purl.obolibrary.org/obo/pr#PR0-short-label>;nf4b2a3a2b8074e59a5e4f5a345615854b1003;<http://www.w3.org/2002/07/owl#Axiom>;"http://proconsortium.org/cgi-bin/entry_pro?id=PR_000031183; <http://purl.obolibrary.org/obo/pr#Gene-based>";TRUE;TRUE;FALSE;FALSE;FALSE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;FALSE;TRUE;FALSE;FALSE

Line 8: http://purl.obolibrary.org/obo/pr#F1Q575_3;http://purl.obolibrary.org/obo/PR_F1Q575;mPGE synthase-2 (human);http://purl.obolibrary.org/obo/PR_Q5XKR9-2;http://purl.obolibrary.org/obo/NCBITaxon_9606;http://purl.obolibrary.org/obo/PR_Q60653;PRO:DNx;"<http://www.ncbi.nlm.nih.gov/gene/1164905>; FATP-2; SUN domain-containing protein 4 (Arabidopsis thaliana)";<http://www.w3.org/2002/07/owl#Class>;http://purl.obolibrary.org/obo/PR_000057376;http://purl.obolibrary.org/obo/PR_000070511;http://purl.obolibrary.org/obo/PR_P0C5K4;"M1 (Influenza A virus (strain A/Udorn/307/1972 H3N2)); PRO:DNx";A protein that is a translation product of the human OSGEPL1 gene or a 1:1 ortholog thereof.;"<http://purl.obolibrary.org/obo/pr#F1Q575>; <http://www.w3.org/2002/07/owl#Class>";FALSE;TRUE;FALSE;FALSE;FALSE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;TRUE;FALSE

Line 9: http://purl.obolibrary.org/obo/pr#Q9SK50_5;http://purl.obolibrary.org/obo/PR_Q9SK50;<http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>;"Category=gene.; UniProtKB:P38070";<https://bar.utoronto.ca/thalemine/porta1.do?externalids=AT4G32470>;"http://purl.obolibrary.org/obo/R0_0002160; <http://www.w3.org/1999/02/22-rdf-syntax-ns#nil>";"http://purl.obolibrary.org/obo/NCBITaxon_10090; http://purl.obolibrary.org/obo/PR_P70658-1; M1 (Influenza A virus (strain A/Udorn/307/1972 H3N2));mCLP1;"http://purl.obolibrary.org/obo/PR_Q6AYZ1; nf002a7abb4e346c0bee6795be1f50998b2242; nf002a7abb4e346c0bee67

```

95be1f50998b3678";"http://purl.obolibrary.org/obo/PR_000065167; CAALFM_C1123
60CA";http://purl.obolibrary.org/obo/PR_Q8K0D0;gene;"Aqr (mouse); nf434fd27b
84b4345befbd5526f317f54b1061";"A protein coding gene ipo8 in zebrafish.; n4c
641700d6bb46049e644d5ff510cf6cb3402";"http://purl.obolibrary.org/obo/pr#Q9SK
50; http://www.w3.org/2002/07/owl#Class";TRUE;FALSE;TRUE;FALSE;FALSE;FALSE;T
RUE;FALSE;FALSE;FALSE;TRUE;TRUE;TRUE;FALSE
Line 10: http://purl.obolibrary.org/obo/pr#000029949_3;http://purl.obolibrar
y.org/obo/PR_000029949;http://purl.obolibrary.org/obo/PR_Q60653;nd7f01bd2140
640c5b620ac65dcb0d808b1459;n6b50d633409242829b2df002a77ad435b569;http://pur
l.obolibrary.org/obo/PR_000008220;"http://purl.obolibrary.org/obo/PR_D3ZUG8;
http://purl.obolibrary.org/obo/PR_Q84LK0";M1 (Influenza A virus (strain A/Ud
orn/307/1972 H3N2));"NVP-2 (rat); Rab11-FIP3 (human)";http://www.genenames.o
rg/cgi-bin/gene_symbol_report?hgnc_id=9863;http://www.geneontology.org/forma
ts/oboInOwl#hasRelatedSynonym;http://purl.obolibrary.org/obo/PR_D4AE48;htt
p://purl.obolibrary.org/obo/PR_P07283;SUN domain-containing protein 4 (Arabi
dopsis thaliana);"https://proconsortium.org/cgi-bin/entry_pro?id=PR_00002994
9; http://purl.obolibrary.org/obo/pr#PR0-short-label";FALSE;FALSE;TRUE;TRUE;
FALSE;FALSE;TRUE;TRUE;FALSE;FALSE;FALSE;FALSE;TRUE;FALSE

```

An exception has occurred, use %tb to see the full traceback.

SystemExit: Terminating due to CSV parsing error.

```

In [10]: import pandas as pd

# Path to your CSV file
csv_path = "10_rows.csv"

try:
    # Read CSV with semicolon delimiter
    df = pd.read_csv(csv_path, delimiter=';')
    print("CSV loaded successfully.")
    print(df.head()) # Show first few rows
except Exception as e:
    print(f"Failed to read CSV: {e}")

```

CSV loaded successfully.

```
Protein URI \
0 http://purl.obolibrary.org/obo/pr#Q8BP67_3
1 http://purl.obolibrary.org/obo/pr#Q9BRT8-4_5
2 http://purl.obolibrary.org/obo/pr#P31362_5
3 http://purl.obolibrary.org/obo/pr#Q0V869-1_2
4 http://purl.obolibrary.org/obo/pr#Q63449-2_5
```

```
Equivalent URI \
0 http://purl.obolibrary.org/obo/PR_Q8BP67
1 http://purl.obolibrary.org/obo/PR_Q9BRT8-4
2 http://purl.obolibrary.org/obo/PR_P31362
3 http://purl.obolibrary.org/obo/PR_Q0V869-1
4 http://purl.obolibrary.org/obo/PR_Q63449-2
```

```
owl:annotatedProperty \
0 hPBRM1/iso:h4
1 http://www.geneontology.org/formats/oboInOwl#h...
2 http://www.pombase.org/spombe/result/SPCC16C4.17
3 PRO:DNx
4 http://purl.obolibrary.org/obo/NCBITaxon_10116
```

```
owl:annotatedSource \
0 http://purl.obolibrary.org/obo/PR_P39361
1 Reactome:R-HSA-6797641
2 Lirf
3 https://bar.utoronto.ca/thalemine/portal.do?ex...
4 Kif20a
```

```
rdfs:comment \
0 At2g17036
1 http://purl.obolibrary.org/obo/R0_0002160; gene
2 http://www.w3.org/2002/07/owl#Axiom; ndbe7d537...
3 n97fa82eb252747c487600ada9ed0f9bab4154
4 n834d695861f34315918db9bc48268214b2037
```

```
oboInOwl:hasBroadSynonym \
0 A protein coding gene acrA in Dictyostelium di...
1 An interleukin-1 receptor type 1 that is a tra...
2 http://purl.obolibrary.org/obo/PR_Q6XPS3-2
3 http://www.geneontology.org/formats/oboInOwl#h...
4 http://purl.obolibrary.org/obo/R0_0002160; An ...
```

```
oboInOwl:hasDbXref \
0 http://www.w3.org/2002/07/owl#Axiom
1 gene
2 ASMT; n2bccccf89d1394cd0a28ac430d42e7938b1348
3 ne12566163ccc45b3833f59b063a82d9db838
4 PMID:8687384
```

```
oboInOwl:hasExactSynonym \
0 http://www.geneontology.org/formats/oboInOwl#h...
1 mCLP1
2 mCLP1
3 n0c1c89fe034a4c3dbc83432e2671820ab2254
4 A proliferation marker protein Ki-67 (human) t...
```

	oboInOwl:hasNarrowSynonym	\
0	http://purl.obolibrary.org/obo/PR_000022733; h...	
1	http://purl.obolibrary.org/obo/PR_P0ACL5; syst...	
2	http://purl.obolibrary.org/obo/PR_G3MU46-2	
3	gene	
4	http://purl.obolibrary.org/obo/PR_P12367; PMID...	

	oboInOwl:hasRelatedSynonym	...	\
0	na2591bfefa654fb8a4039152cf663b4eb2431	...	
1	gene	...	
2	http://www.w3.org/2002/07/owl#Axiom; nc3fe8eaf...	...	
3	n2deaf7d8b8f54dbbabb317f2c29f7865b2281	...	
4	http://www.w3.org/2002/07/owl#Axiom	...	

	oboInOwl:hasBroadSynonym_uri_valid	oboInOwl:hasDbXref_uri_valid	\
0	True	False	
1	True	True	
2	False	True	
3	False	True	
4	False	True	

	oboInOwl:hasExactSynonym_uri_valid	oboInOwl:hasNarrowSynonym_uri_valid	\
0	False	False	
1	True	False	
2	True	False	
3	True	True	
4	True	False	

	oboInOwl:hasRelatedSynonym_uri_valid	rdfls:label_uri_valid	\
0	True	False	
1	True	False	
2	False	False	
3	True	False	
4	False	True	

	owl:onProperty_uri_valid	owl:someValuesFrom_uri_valid	\
0	False	True	
1	True	True	
2	False	True	
3	True	True	
4	False	True	

	rdfls:subClassOf_uri_valid	rdf:type_uri_valid
0	True	False
1	True	False
2	True	False
3	True	False
4	True	False

[5 rows x 29 columns]

```
In [12]: import pandas as pd
import requests
from requests.exceptions import RequestException
```

```

# Load your CSV (adjust filename and delimiter as needed)
df = pd.read_csv('10_rows.csv', delimiter=';') # <-- add delimiter here

# List of columns that contain URLs you want to fetch and replace with page
url_columns = [
    'Protein URI',
    'Equivalent URI',
    'owl:annotatedProperty',
    'owl:annotatedSource',
    'rdfs:comment',
    'oboInOwl:hasBroadSynonym',
    'oboInOwl:hasDbXref',
    'oboInOwl:hasExactSynonym',
    'oboInOwl:hasNarrowSynonym',
    'oboInOwl:hasRelatedSynonym',
    # add any other columns you want to fetch page content for
]

def fetch_url_content(url):
    try:
        if pd.isna(url) or not isinstance(url, str) or not url.startswith('h
            return url # return as is if not a valid HTTP URL
        response = requests.get(url, timeout=5)
        response.raise_for_status()
        return response.text # return page content (HTML or text)
    except RequestException as e:
        print(f"Failed to fetch {url}: {e}")
        return url # fallback to original URL if fetch fails

# Apply fetching function to each URL cell in the columns
for col in url_columns:
    if col in df.columns:
        print(f"Processing column: {col}")
        df[col] = df[col].apply(fetch_url_content)

# Save the updated DataFrame to a new CSV
df.to_csv('updated_with_page_content.csv', index=False)

print("New CSV with page content saved as 'updated_with_page_content.csv'")

```

Processing column: Protein URI
Processing column: Equivalent URI
Processing column: owl:annotatedProperty
Failed to fetch http://purl.obolibrary.org/obo/NCBITaxon_10116: HTTPSConnectionPool(host='www.ontobee.org', port=443): Read timed out. (read timeout=5)
Processing column: owl:annotatedSource
Processing column: rdfs:comment
Failed to fetch http://purl.obolibrary.org/obo/RO_0002160; gene: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/RO_0002160;%20gene
Failed to fetch http://purl.obolibrary.org/obo/NCBITaxon_9606: HTTPSConnectionPool(host='www.ontobee.org', port=443): Read timed out. (read timeout=5)
Failed to fetch http://purl.obolibrary.org/obo/IAO_0000115: HTTPSConnectionPool(host='www.ontobee.org', port=443): Read timed out. (read timeout=5)
Processing column: oboInOwl:hasBroadSynonym
Failed to fetch http://purl.obolibrary.org/obo/RO_0002160; An interleukin-1 receptor type 1 that is a translation product of some mRNA whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by UniProtKB:P13504-1 or a 1:1 ortholog thereof.; UniProtKB:Q7ZUP1: 403 Client Error: Forbidden for url: http://purl.obolibrary.org/obo/RO_0002160;%20An%20interleukin-1%20receptor%20type%201%20that%20is%20a%20translation%20product%20of%20some%20mRNA%20whose%20exon%20structure%20and%20start%20site%20selection%20renders%20it%20capable%20of%20giving%20rise%20to%20a%20protein%20with%20the%20amino%20acid%20sequence%20represented%20by%20UniProtKB:P13504-1%20or%20a%201:1%20ortholog%20thereof.;%20UniProtKB:Q7ZUP1
Failed to fetch http://purl.obolibrary.org/obo/RO_0002160; http://www.w3.org/1999/02/22-rdf-syntax-ns#nil: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/RO_0002160;%20http://www.w3.org/1999/02/22-rdf-syntax-ns#nil
Processing column: oboInOwl:hasDbXref
Failed to fetch http://purl.obolibrary.org/obo/NCBITaxon_10090; http://purl.obolibrary.org/obo/PR_P70658-1; M1 (Influenza A virus (strain A/Udorn/307/1972 H3N2)): 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/NCBITaxon_10090;%20http://purl.obolibrary.org/obo/PR_P70658-1;%20M1%20(Influenza%20A%20virus%20(strain%20A/Udorn/307/1972%20H3N2))
Failed to fetch http://purl.obolibrary.org/obo/PR_D3ZUG8; http://purl.obolibrary.org/obo/PR_Q84LK0: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_D3ZUG8;%20http://purl.obolibrary.org/obo/PR_Q84LK0
Processing column: oboInOwl:hasExactSynonym
Failed to fetch http://www.ncbi.nlm.nih.gov/gene/1164905; FATP-2; SUN domain-containing protein 4 (Arabidopsis thaliana): 404 Client Error: Not Found for url: https://www.ncbi.nlm.nih.gov/gene/1164905;%20FATP-2;%20SUN%20domain-containing%20protein%204%20(Arabidopsis%20thaliana)
Processing column: oboInOwl:hasNarrowSynonym
Failed to fetch http://purl.obolibrary.org/obo/PR_000022733; http://purl.obolibrary.org/obo/PR_Q9NVE5-2; PR:Q23490: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_000022733;%20http://purl.obolibrary.org/obo/PR_Q9NVE5-2;%20PR:Q23490
Failed to fetch http://purl.obolibrary.org/obo/PR_P0ACL5; system B(0) neutral amino acid transporter AT1 (human): 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_P0ACL5;%20system%20B(0)%20neutral%20amino%20acid%20transporter%20AT1%20(human)
Failed to fetch http://purl.obolibrary.org/obo/PR_P12367; PMID:6190803: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_P12367;%20PMID:6190803

Failed to fetch http://purl.obolibrary.org/obo/PR_Q6AYZ1; nf002a7abb4e346c0bee6795be1f50998b2242; nf002a7abb4e346c0bee6795be1f50998b3678: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678

Processing column: oboInOwl:hasRelatedSynonym

Failed to fetch http://purl.obolibrary.org/obo/PR_000065167; CAALFM_C112360CA: 404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_000065167;%20CAALFM_C112360CA

Failed to fetch http://www.genenames.org/cgi-bin/gene_symbol_report?hgnc_id=9863: 404 Client Error: Not Found for url: https://www.genenames.org:443/cgi-bin/gene_symbol_report?hgnc_id=9863

New CSV with page content saved as 'updated_with_page_content.csv'

```
In [16]: import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

# Data as provided (reformatted for readability in code)
data = [
    ["owl:annotatedProperty", "Timeout", "http://purl.obolibrary.org/obo/NCE",
     "HTTPSConnectionPool(host='www.ontobee.org', port=443): Read timed out.",
     "owl:annotatedSource", "", "", "Processing successful, no error listed"],
    ["rdfs:comment", "Not Found (404)", "http://purl.obolibrary.org/obo/R0_0",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     "rdfs:comment", "Timeout", "http://purl.obolibrary.org/obo/NCBITaxon_9606",
     "HTTPSConnectionPool(host='www.ontobee.org', port=443): Read timed out.",
     "rdfs:comment", "Timeout", "http://purl.obolibrary.org/obo/IAO_0000115",
     "HTTPSConnectionPool(host='www.ontobee.org', port=443): Read timed out.",
     ["oboInOwl:hasBroadSynonym", "Forbidden (403)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     "403 Client Error: Forbidden for url: http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasBroadSynonym", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasDbXref", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasDbXref", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasExactSynonym", "Not Found (404)", "http://www.ncbi.nlm.nih.gov/gene/9863",
     "404 Client Error: Not Found for url: https://www.ncbi.nlm.nih.gov/gene/9863",
     ["oboInOwl:hasNarrowSynonym", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasNarrowSynonym", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasNarrowSynonym", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasNarrowSynonym", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;nf002a7abb4e346c0bee6795be1f50998b2242;nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasRelatedSynonym", "Not Found (404)", "http://purl.obolibrary.org/obo/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     "404 Client Error: Not Found for url: http://www.berkeleybop.org/ontologies/PR_Q6AYZ1;%20nf002a7abb4e346c0bee6795be1f50998b2242;%20nf002a7abb4e346c0bee6795be1f50998b3678",
     ["oboInOwl:hasRelatedSynonym", "Not Found (404)", "http://www.genenames.org:443/cgi-bin/gene_symbol_report?hgnc_id=9863",
     "404 Client Error: Not Found for url: https://www.genenames.org:443/cgi-bin/gene_symbol_report?hgnc_id=9863"],
    ]
]

# Create DataFrame
df = pd.DataFrame(data, columns=["Column", "Error Type", "URL", "Details"])
```



```

# Set figure size for image
fig, ax = plt.subplots(figsize=(18, 8))

# Hide axes
ax.xaxis.set_visible(False)
ax.yaxis.set_visible(False)
ax.set_frame_on(False)

# Create a color map for Error Type highlighting
color_map = {
    "Timeout": "#f9d5d3",
    "Not Found (404)": "#fcebbd",
    "Forbidden (403)": "#f4c2c2",
    "": "#d3f9d8", # No error / success - greenish
}

# Apply color to rows based on Error Type
row_colors = df["Error Type"].map(lambda x: color_map.get(x, "#ffffff"))

# Draw the table
table = ax.table(cellText=df.values,
                 collabels=df.columns,
                 cellColours=[[row_colors[i]]*4 for i in range(len(df))],
                 colColours=["#2f4f4f"]*4,
                 cellLoc='left',
                 loc='center')

table.auto_set_font_size(False)
table.set_fontsize(10)
table.auto_set_column_width([0, 1, 2, 3])

# Style header cells
for (row, col), cell in table.get_celld().items():
    if row == 0:
        cell.set_text_props(weight='bold', color='white')
        cell.set_facecolor("#2f4f4f")

# Adjust layout
plt.tight_layout()

# Save as image file
img_path = "beautiful_table.png"
plt.savefig(img_path, dpi=150)
plt.close()

img_path

```

Out[16]: 'beautiful_table.png'

```

In [21]: import pandas as pd
import openai

# Set Groq/OpenAI API credentials
openai.api_key = "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"
openai.api_base = "https://api.groq.com/openai/v1"

```

```

# Load the CSVs with semicolon delimiter (based on your example)
uri_df = pd.read_csv("10_rows.csv", delimiter=';')
content_df = pd.read_csv("updated_with_page_content.csv", delimiter=';')

# Ensure same shape
assert uri_df.shape == content_df.shape, "Files must have the same shape"

# Optional: Limit number of rows to sample for quicker testing
NUM_SAMPLES = 100
if len(uri_df) > NUM_SAMPLES:
    sample_indices = uri_df.sample(NUM_SAMPLES, random_state=42).index
    sample_df = uri_df.loc[sample_indices]
    sample_df_content = content_df.loc[sample_indices]
else:
    sample_df = uri_df.copy()
    sample_df_content = content_df.copy()

# Columns to exclude from evaluation (keep as is)
exclude_columns = ["Protein URI", "Equivalent URI", "owl:annotatedProperty",

results = []

# Iterate through rows and columns
for idx in sample_df.index:
    for col in sample_df.columns:
        if col in exclude_columns:
            continue

        uri_value = str(sample_df.at[idx, col]).strip()
        content_value = str(sample_df_content.at[idx, col]).strip()

        # Skip empty or 'nan' content values
        if not content_value or content_value.lower() == "nan":
            continue

        # Compose prompt for the LLM
        prompt = (
            f"Answer 'Yes' or 'No' only.\n"
            f"Is the following content relevant to the property '{col}' "
            f"of the entity identified by the URI: '{uri_value}'?\n\n"
            f"Content: '{content_value}'"
        )

        try:
            response = openai.ChatCompletion.create(
                model="meta-llama/llama-4-scout-17b-16e-instruct",
                messages=[{"role": "user", "content": prompt}],
                temperature=0.0,
                max_tokens=5,
            )

            # Parse the LLM answer (take first line, normalize to Yes/No)
            raw_answer = response.choices[0].message["content"].strip().split()
            answer = "Yes" if raw_answer[0].lower().startswith("yes") else "No"

            results.append({

```

```

        "Entity URI": uri_value,
        "Property": col,
        "Content": content_value,
        "Prompt": prompt,
        "Relevant": answer
    })

    except Exception as e:
        print(f"[Error] Row {idx}, Column {col}: {e}")

# Save all results to CSV
results_df = pd.DataFrame(results)
results_df.to_csv("content_relevance_results.csv", index=False, sep=';')

print("✅ Done! Results saved to content_relevance_results.csv")

```

```

-----
ParserError                                Traceback (most recent call last)
Cell In[21], line 10
      8 # Load the CSVs with semicolon delimiter (based on your example)
      9 uri_df = pd.read_csv("10_rows.csv", delimiter=';')
--> 10 content_df = pd.read_csv("updated_with_page_content.csv", delimiter
    =';')
      12 # Ensure same shape
      13 assert uri_df.shape == content_df.shape, "Files must have the same s
hape"

File /opt/anaconda3/lib/python3.12/site-packages/pandas/io/parsers/readers.p
y:1026, in read_csv(filepath_or_buffer, sep, delimiter, header, names, index
_col, usecols, dtype, engine, converters, true_values, false_values, skipini
tialspace, skiprows, skipfooter, nrows, na_values, keep_default_na, na_filt
er, verbose, skip_blank_lines, parse_dates, infer_datetime_format, keep_date
_col, date_parser, date_format, dayfirst, cache_dates, iterator, chunksize, c
ompression, thousands, decimal, lineterminator, quotechar, quoting, doublequ
ote, escapechar, comment, encoding, encoding_errors, dialect, on_bad_lines,
delim_whitespace, low_memory, memory_map, float_precision, storage_options,
dtype_backend)
    1013 kwds_defaults = _refine_defaults_read(
    1014     dialect,
    1015     delimiter,
    (...)
    1022     dtype_backend=dtype_backend,
    1023 )
    1024 kwds.update(kwds_defaults)
-> 1026 return _read(filepath_or_buffer, kwds)

File /opt/anaconda3/lib/python3.12/site-packages/pandas/io/parsers/readers.p
y:626, in _read(filepath_or_buffer, kwds)
    623     return parser
    625 with parser:
--> 626     return parser.read(nrows)

File /opt/anaconda3/lib/python3.12/site-packages/pandas/io/parsers/readers.p
y:1923, in TextFileReader.read(self, nrows)
    1916 nrows = validate_integer("nrows", nrows)
    1917 try:
    1918     # error: "ParserBase" has no attribute "read"
    1919     (
    1920         index,
    1921         columns,
    1922         col_dict,
-> 1923     ) = self._engine.read( # type: ignore[attr-defined]
    1924         nrows
    1925     )
    1926 except Exception:
    1927     self.close()

File /opt/anaconda3/lib/python3.12/site-packages/pandas/io/parsers/c_parser_
wrapper.py:234, in CParserWrapper.read(self, nrows)
    232 try:
    233     if self.low_memory:
--> 234         chunks = self._reader.read_low_memory(nrows)

```

```

235         # destructive to chunks
236         data = _concatenate_chunks(chunks)

File parsers.pyx:838, in pandas._libs.parsers.TextReader.read_low_memory()
File parsers.pyx:905, in pandas._libs.parsers.TextReader._read_rows()
File parsers.pyx:874, in pandas._libs.parsers.TextReader._tokenize_rows()
File parsers.pyx:891, in pandas._libs.parsers.TextReader._check_tokenize_status()
File parsers.pyx:2061, in pandas._libs.parsers.raise_parser_error()

ParserError: Error tokenizing data. C error: Expected 1 fields in line 28, saw 2

```

```

In [23]: with open("updated_with_page_content.csv", encoding='utf-8') as f:
        header = next(f).strip().split(",")

        records = []
        current_record_lines = []
        for line in f:
            if line.strip() == "": # blank line separates records
                if current_record_lines:
                    records.append("".join(current_record_lines).strip())
                    current_record_lines = []
            else:
                current_record_lines.append(line)
        if current_record_lines:
            records.append("".join(current_record_lines).strip())

        print(f"Parsed {len(records)} records after header.")
        print("Preview first record:")
        print(records[0][:1000])

```

```

Parsed 3937 records after header.
Preview first record:
"<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<HTML>
<HEAD><TITLE>Protein Ontology </TITLE>
<LINK href="/styles.css" type="text/css" rel="stylesheet">
<link rel="shortcut icon" href="favicon.ico">
</HEAD>

```

```

In [24]: import pandas as pd

        with open("updated_with_page_content.csv", encoding='utf-8') as f:
            header = next(f).strip().split(",")

            records = []
            current_record_lines = []
            for line in f:
                if line.strip() == "": # blank line separates records
                    if current_record_lines:
                        records.append("".join(current_record_lines).strip())

```

```

        current_record_lines = []
    else:
        current_record_lines.append(line)
    if current_record_lines:
        records.append("".join(current_record_lines).strip())

# Now create DataFrame with a column named after the first header column
df = pd.DataFrame(records, columns=[header[0]])

print(df.head())
print(f"DataFrame shape: {df.shape}")

```

```

                                Protein URI
0  "<!DOCTYPE HTML PUBLIC ""-//W3C//DTD HTML 4.01...
1  <BODY bgColor=""#cccccc"" background=""/images...
2  <table border=""0"" cellspacing=""0"" cellpadd...
3  <TABLE cellSpacing=0 cellPadding=0 width=100% ...
4                                     <TR>
DataFrame shape: (3937, 1)

```

```

In [27]: import pandas as pd

# Read the file line-by-line, skip header, then join all lines into one string
with open("updated_with_page_content.csv", encoding='utf-8') as f:
    lines = f.readlines()

header = lines[0]
html_lines = lines[1:] # all lines after header

# Join all html lines into a single string (strip leading tabs if needed)
html_content = "".join(line.lstrip("\t") for line in html_lines)

# Now build a DataFrame with one row containing that html content under the
df = pd.DataFrame({
    "Protein URI": [html_content]
})

print(df.shape)
print(df.iloc[0,0][:1000]) # print first 1000 chars to check

```

```

(1, 1)
"<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<HTML>
<HEAD><TITLE>Protein Ontology </TITLE>
<LINK href= ""/styles.css"" type= ""text/css"" rel= ""stylesheet"">
<link rel= ""shortcut icon"" href= ""favicon.ico"">
</HEAD>

<BODY bgColor= ""#cccccc"" background= ""/images/temp/bg02.gif"" leftMargin
= ""0"" topMargin= ""0"" marginwidth= ""0"" marginheight= ""0"" >

<table border= ""0"" cellspacing= ""0"" cellpadding= ""0"" width=100%>
<tr><td>

<TABLE cellSpacing=0 cellPadding=0 width=100% background=bannerBg.png border
=0 >
<TR><TD colspan=8><img src= '/images/empty_line.GIF' width= '1' height= '9'></T
D></TR>
<TR><td bgcolor= #EEEEEE colspan=8 height=1 nowrap><img src= '/images/empty_li
ne.GIF' width= '1' height= '3'></td></TR>
<tr>
<td><img src= '/images/empty_line.GIF' width= '8' height= '76'></td>
<td align= center ><img
src= '/images/empty_line.GIF' width= '1' height= '3'><br><img
src= '/images/empty_line.GIF' width= '3' height= '1'><a
href= /pro.shtml><img
alt= 'back to PRO homepage' src= '/PR

```

```

In [32]: import pandas as pd
from bs4 import BeautifulSoup

# Load CSV file
df = pd.read_csv('updated_with_page_content.csv')

# Function to extract plain text from HTML
def extract_text_from_html(html):
    if pd.isna(html):
        return ""
    soup = BeautifulSoup(html, 'html.parser')
    return soup.get_text(separator=' ', strip=True)

# Function to extract <title> content from HTML (optional)
def extract_title(html):
    if pd.isna(html):
        return ""
    soup = BeautifulSoup(html, 'html.parser')
    title_tag = soup.title
    return title_tag.string if title_tag else ""

# Columns to clean from HTML content
html_columns = [
    'rdfs:comment',
    'oboInOwl:hasBroadSynonym',
    'oboInOwl:hasExactSynonym',
    'oboInOwl:hasNarrowSynonym',
    'oboInOwl:hasRelatedSynonym',

```

```

    'rdfs:label'
]

# Clean HTML content in those columns
for col in html_columns:
    if col in df.columns:
        df[col] = df[col].apply(extract_text_from_html)

# Optionally extract titles from a specific column (example: 'rdfs:comment')
if 'rdfs:comment' in df.columns:
    df['comment_title'] = df['rdfs:comment'].apply(extract_title)

# Save cleaned DataFrame to new CSV
df.to_csv('output_cleaned.csv', index=False)

```

/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_2485/1731697583.py:11: MarkupResemblesLocatorWarning: The input looks more like a filename than a markup. You may want to open this file and pass the filehandle into BeautifulSoup.

```
soup = BeautifulSoup(html, 'html.parser')
```

/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_2485/1731697583.py:11: MarkupResemblesLocatorWarning: The input looks more like a URL than a markup. You may want to use an HTTP client like requests to get the document behind the URL, and feed that document to BeautifulSoup.

```
soup = BeautifulSoup(html, 'html.parser')
```

/opt/anaconda3/lib/python3.12/html/parser.py:171: XMLParsedAsHTMLWarning: It looks like you're parsing an XML document using an HTML parser. If this really is an HTML document (maybe it's XHTML?), you can ignore or filter this warning. If it's XML, you should know that using an XML parser will be more reliable. To parse this document as XML, make sure you have the lxml package installed, and pass the keyword argument `features="xml"` into the BeautifulSoup constructor.

```
k = self.parse_starttag(i)
```

/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_2485/1731697583.py:18: MarkupResemblesLocatorWarning: The input looks more like a filename than a markup. You may want to open this file and pass the filehandle into BeautifulSoup.

```
soup = BeautifulSoup(html, 'html.parser')
```

/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_2485/1731697583.py:18: MarkupResemblesLocatorWarning: The input looks more like a URL than a markup. You may want to use an HTTP client like requests to get the document behind the URL, and feed that document to BeautifulSoup.

```
soup = BeautifulSoup(html, 'html.parser')
```

```

In [33]: import pandas as pd
        from bs4 import BeautifulSoup

# Load your CSV
df = pd.read_csv('updated_with_page_content.csv')

# Function to extract text content from HTML
def extract_text_from_html(html):
    if pd.isna(html):
        return ""
    soup = BeautifulSoup(html, 'html.parser')
    return soup.get_text(separator=' ', strip=True)

```



```

# List of columns with HTML content to clean
html_columns = [
    'rdfs:comment',
    'oboInOwl:hasBroadSynonym',
    'oboInOwl:hasExactSynonym',
    'oboInOwl:hasNarrowSynonym',
    'oboInOwl:hasRelatedSynonym',
    'rdfs:label'
]

# Apply the extraction to each relevant column
for col in html_columns:
    if col in df.columns:
        df[col] = df[col].apply(extract_text_from_html)

# Save the cleaned DataFrame
df.to_csv('output_cleaned.csv', index=False)

```

/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_2485/630647825.py:11: MarkupResemblesLocatorWarning: The input looks more like a filename than a markup. You may want to open this file and pass the filehandle into BeautifulSoup.

soup = BeautifulSoup(html, 'html.parser')

/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_2485/630647825.py:11: MarkupResemblesLocatorWarning: The input looks more like a URL than a markup. You may want to use an HTTP client like requests to get the document behind the URL, and feed that document to BeautifulSoup.

soup = BeautifulSoup(html, 'html.parser')

/opt/anaconda3/lib/python3.12/html/parser.py:171: XMLParsedAsHTMLWarning: It looks like you're parsing an XML document using an HTML parser. If this really is an HTML document (maybe it's XHTML?), you can ignore or filter this warning. If it's XML, you should know that using an XML parser will be more reliable. To parse this document as XML, make sure you have the lxml package installed, and pass the keyword argument `features="xml"` into the BeautifulSoup constructor.

k = self.parse_starttag(i)

In [34]: **import** pandas **as** pd

```

# Read the cleaned CSV file
df_cleaned = pd.read_csv('output_cleaned.csv')

# Now df_cleaned contains your cleaned data
print(df_cleaned.head()) # Display first few rows to verify

```

```

Protein URI \
0 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 T...
1 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 T...
2 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 T...
3 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 T...
4 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 T...

Equivalent URI \
0 \t\t<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4....
1 \t\t<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4....
2 \t\t<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4....
3 \t\t<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4....
4 \t\t<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4....

owl:annotatedProperty \
0 hpBRM1/iso:h4
1 <?xml version="1.0"?>\n<rdf:RDF xmlns="http://...
2 <!doctype html>\n<html data-beasties-container...
3 PRO:DNx
4 http://purl.obolibrary.org/obo/NCBITaxon_10116

owl:annotatedSource \
0 \t\t<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4....
1 Reactome:R-HSA-6797641
2 Lirf
3 <!DOCTYPE html>\n\n<!-- layout.jsp -->\n<html ...
4 Kif20a

rdfs:comment \
0 At2g17036
1 http://purl.obolibrary.org/obo/R0_0002160; gene
2 @prefix dc: .\r\n@prefix grddl: .\r\n@prefix o...
3 n97fa82eb252747c487600ada9ed0f9bab4154
4 n834d695861f34315918db9bc48268214b2037

oboInOwl:hasBroadSynonym \
0 A protein coding gene acrA in Dictyostelium di...
1 An interleukin-1 receptor type 1 that is a tra...
2 Protein Ontology Home | Linked Open Data | Bro...
3 OBO Format metamodel. This meta-ontology is se...
4 http://purl.obolibrary.org/obo/R0_0002160; An ...

oboInOwl:hasDbXref \
0 @prefix dc: <http://purl.org/dc/elements/1.1/>...
1 gene
2 ASMT; n2bccccf89d1394cd0a28ac430d42e7938b1348
3 ne12566163ccc45b3833f59b063a82d9db838
4 PMID:8687384

oboInOwl:hasExactSynonym \
0 OBO Format metamodel. This meta-ontology is se...
1 mCLP1
2 mCLP1
3 n0c1c89fe034a4c3dbc83432e2671820ab2254
4 A proliferation marker protein Ki-67 (human) t...

```

```

                                oboInOwl:hasNarrowSynonym \
0 http://purl.obolibrary.org/obo/PR_000022733; h...
1 http://purl.obolibrary.org/obo/PR_P0ACL5; syst...
2 Protein Ontology Home | Linked Open Data | Bro...
3 gene
4 http://purl.obolibrary.org/obo/PR_P12367; PMID...

                                oboInOwl:hasRelatedSynonym ... \
0 na2591bfefa654fb8a4039152cf663b4eb2431 ...
1 gene ...
2 @prefix dc: .\r\n@prefix grddl: .\r\n@prefix o... ...
3 n2deaf7d8b8f54dbbabd317f2c29f7865b2281 ...
4 @prefix dc: .\r\n@prefix grddl: .\r\n@prefix o... ...

oboInOwl:hasBroadSynonym_uri_valid oboInOwl:hasDbXref_uri_valid \
0 True False
1 True True
2 False True
3 False True
4 False True

oboInOwl:hasExactSynonym_uri_valid oboInOwl:hasNarrowSynonym_uri_valid \
0 False False
1 True False
2 True False
3 True True
4 True False

oboInOwl:hasRelatedSynonym_uri_valid rdfs:label_uri_valid \
0 True False
1 True False
2 False False
3 True False
4 False True

owl:onProperty_uri_valid owl:someValuesFrom_uri_valid \
0 False True
1 True True
2 False True
3 True True
4 False True

rdfs:subClassOf_uri_valid rdf:type_uri_valid
0 True False
1 True False
2 True False
3 True False
4 True False

```

[5 rows x 29 columns]

```

In [36]: import pandas as pd

# Load cleaned CSV
df = pd.read_csv('output_cleaned.csv')

```

```

def is_related(cell_value, column_name):
    # Handle missing or empty cells
    if cell_value is None:
        return "no"

    # Convert everything to string safely
    cell_str = str(cell_value)

    # If it is a boolean string like 'True' or 'False', consider no
    if cell_str.lower() in ['true', 'false']:
        return "no"

    if cell_str.strip() == "":
        return "no"

    # Normalize and split column name by common delimiters
    keywords = column_name.lower().replace(':', ' ').replace('_', ' ').split()
    cell_text = cell_str.lower()

    for kw in keywords:
        if kw in cell_text:
            return "yes"

    return "no"

# Create a new DataFrame with yes/no values
df_related = df.copy()
for col in df.columns:
    df_related[col] = df[col].apply(lambda x: is_related(x, col))

# Save the yes/no DataFrame
df_related.to_csv('output_related_check.csv', index=False)

print("Done. Saved to output_related_check.csv")

```

Done. Saved to output_related_check.csv

```

In [38]: import pandas as pd
import openai
import time

# Set Groq/OpenAI API credentials
openai.api_key = "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"
openai.api_base = "https://api.groq.com/openai/v1"

# Load cleaned CSV
df = pd.read_csv('output_cleaned.csv')

def check_related_openai(cell_value, column_name):
    # Handle empty or NaN cells
    if pd.isna(cell_value) or str(cell_value).strip() == "":
        return "no"

    prompt = f"Is the following text related to the column name?\n\nColumn r

    try:

```

```

        response = openai.ChatCompletion.create(
            model="meta-llama/llama-4-scout-17b-16e-instruct",
            messages=[
                {"role": "system", "content": "You answer only with 'yes' or 'no'"},
                {"role": "user", "content": prompt}
            ],
            temperature=0
        )
        answer = response.choices[0].message.content.strip().lower()
        if "yes" in answer:
            return "yes"
        else:
            return "no"
    except Exception as e:
        print(f"OpenAI API error: {e}")
        return "no"

# Process DataFrame columns
df_related = df.copy()
for col in df.columns:
    print(f"Processing column: {col}")
    df_related[col] = df[col].apply(lambda x: check_related_openai(x, col))
    time.sleep(1) # Adjust sleep to avoid rate limits

# Save results
df_related.to_csv('output_related_check_openai.csv', index=False)
print("Done. Saved to output_related_check_openai.csv")

```

Processing column: Protein URI

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28872, Requested 3716. Please try again in 5.176s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28852, Requested 3716. Please try again in 5.135s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28831, Requested 3716. Please try again in 5.093s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

Processing column: Equivalent URI

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28308, Requested 3474. Please try again in 3.564s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28265, Requested 3404. Please try again in 3.338s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28245, Requested 3515. Please try again in 3.52s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28225, Requested 3489. Please try again in 3.428s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28203, Requested 3430. Please try again in 3.265s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28180, Requested 7686. Please try again in 11.731s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28160, Requested 3504. Please try again in 3.328s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28140, Requested 3502. Please try again in 3.284s. Need more tokens? Upgrade to Dev Tier today

at <https://console.groq.com/settings/billing>
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28107, Requested 18186. Please try again in 32.586s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28086, Requested 4045. Please try again in 4.261999999s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
Processing column: owl:annotatedProperty
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 27554, Requested 7231. Please try again in 9.57s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 27519, Requested 4376. Please try again in 3.789s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 27505, Requested 7231. Please try again in 9.471s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 27484, Requested 3482. Please try again in 1.931s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 27462, Requested 7231. Please try again in 9.385s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
Processing column: owl:annotatedSource
OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 26938, Requested 3456. Please try again in 788ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
OpenAI API error: Request too large for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Requested 51094, please reduce your message size and try again. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing> {"error":{"message":"Request too large for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Requested 51094, please reduce your message size and try again. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>", "type":"tokens", "code":"rate_limit_exceeded"}}

413 {'error': {'message': 'Request too large for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` ser

vice tier `on\_demand` on tokens per minute (TPM): Limit 30000, Requested 51094, please reduce your message size and try again. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>, 'type': 'tokens', 'code': 'rate_limit_exceeded'}} {'Date': 'Mon, 26 May 2025 08:47:35 GMT', 'Content-Type': 'application/json', 'Content-Length': '409', 'Connection': 'keep-alive', 'Cache-Control': 'private, max-age=0, no-store, no-cache, must-revalidate', 'retry-after': '96', 'vary': 'Origin', 'x-groq-region': 'gcp-europe-west3', 'x-ratelimit-limit-requests': '1000', 'x-ratelimit-limit-tokens': '30000', 'x-ratelimit-remaining-requests': '986', 'x-ratelimit-remaining-tokens': '3125', 'x-ratelimit-reset-requests': '20m9.382999999s', 'x-ratelimit-reset-tokens': '53.75s', 'x-request-id': 'req_01jw5wd78zeeg8qs2cj2qd hgbz', 'x-should-retry': 'false', 'cf-cache-status': 'DYNAMIC', 'Server': 'cloudflare', 'CF-RAY': '945c0234798a0a6d-AMS', 'alt-svc': 'h3=":443"; ma=86400'}

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 26852, Requested 3555. Please try again in 813ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 26831, Requested 3368. Please try again in 398ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 26826, Requested 7230. Please try again in 8.111s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

Processing column: rdfs:comment

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 26306, Requested 5354. Please try again in 3.319s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

Processing column: oboInOwl:hasBroadSynonym

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.866s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.825s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.79s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.754s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.716s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

Processing column: oboInOwl:hasDbXref

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28318, Requested 6182. Please try again in 9s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 614ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 579ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 544ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 506.999999ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 467ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 433ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 395.999999ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 358ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28133, Requested 24739. Please try again in 45.744s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

Processing column: oboIn0wl:hasExactSynonym

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.815s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.779s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.741s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.699s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.661s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.6s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.558s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.52s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 1.394s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

Processing column: oboIn0wl:hasNarrowSynonym

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 343ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 302.999999ms. Need more tokens? Upgrade to Dev Tier today

at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 264.999999ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 225ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 183.999999ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28531, Requested 5357. Please try again in 7.775s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on tokens per minute (TPM): Limit 30000, Used 28510, Requested 5357. Please try again in 7.734s. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 64ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>
 OpenAI API error: Rate limit reached for model `meta-llama/llama-4-scout-17b-16e-instruct` in organization `org\_01jsvaavkcfqd9nka3jca94pr6` service tier `on\_demand` on requests per minute (RPM): Limit 30, Used 30, Requested 1. Please try again in 26.999999ms. Need more tokens? Upgrade to Dev Tier today at <https://console.groq.com/settings/billing>

```
-----
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[38], line 42
      40     print(f"Processing column: {col}")
      41     df_related[col] = df[col].apply(lambda x: check_related_openai
(x, col))
--> 42     time.sleep(1) # Adjust sleep to avoid rate limits
      44 # Save results
      45 df_related.to_csv('output_related_check_openai.csv', index=False)

KeyboardInterrupt:
```

```
In [48]: import pandas as pd
import openai
import time

# Set Groq/OpenAI API credentials
openai.api_key = "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"
openai.api_base = "https://api.groq.com/openai/v1"
```

```

df = pd.read_csv('output_cleaned_renamed.csv')

def chunk_list(lst, chunk_size):
    for i in range(0, len(lst), chunk_size):
        yield lst[i:i + chunk_size]

# def check_batch_openai(texts, column_name):
#     MAX_TEXT_LENGTH = 300 # max chars per text to reduce prompt size
#     texts = [t[:MAX_TEXT_LENGTH] for t in texts]

#     prompt = f"Column name: '{column_name}'\nFor each text below, answer '
#     for i, text in enumerate(texts, 1):
#         prompt += f"{i}. {text}\n"

#     retries = 3
#     for attempt in range(retries):
#         try:
#             response = openai.ChatCompletion.create(
#                 model="meta-llama/llama-4-scout-17b-16e-instruct",
#                 messages=[
#                     {"role": "system", "content": "You answer only with 'y
#                     {"role": "user", "content": prompt}
#                 ],
#                 temperature=0
#             )
#             content = response.choices[0].message.content.strip().lower()

#             answers = []
#             for line in content.splitlines():
#                 if '.' in line:
#                     ans = line.split('.', 1)[1].strip()
#                     if 'yes' in ans:
#                         answers.append('yes')
#                     else:
#                         answers.append('no')

#             if len(answers) == len(texts):
#                 return answers
#             else:
#                 print("Warning: number of answers mismatch, defaulting to
#                 return ['no'] * len(texts)

#         except openai.error.RateLimitError as e:
#             wait_time = (attempt + 1) * 10
#             print(f"Rate limit hit. Waiting {wait_time}s before retry...")
#             time.sleep(wait_time)
#         except Exception as e:
#             print(f"OpenAI API error: {e}")
#             return ['no'] * len(texts)

#     return ['no'] * len(texts)
def check_batch_openai(texts, column_name):
    MAX_TEXT_LENGTH = 300
    texts = [t[:MAX_TEXT_LENGTH] for t in texts]

    # prompt = f"Column name: '{column_name}'\nFor each text below, answer '

```

```

# for i, text in enumerate(texts, 1):
#     prompt += f"{i}. {text}\n"
# prompt = f"Given the column name: '{column_name}', determine if each c
# for i, text in enumerate(texts, 1):
#     prompt += f"{i}. {text}\n"
# prompt = f"Column: '{column_name}'\nIs the following text related to t
# for i, text in enumerate(texts, 1):
#     prompt += f"{i}. {text}\n"
# prompt = f"Column name: '{column_name}'\nFor each of the following tex
# for i, text in enumerate(texts, 1):
#     prompt += f"{i}. {text}\n"
prompt = f"Is '{column_name}' related to each text below? please answer '
for i, text in enumerate(texts, 1):
    prompt += f"{i}. {text}\n"

```

```

retries = 3
for attempt in range(retries):
    try:
        response = openai.ChatCompletion.create(
            model="meta-llama/llama-4-scout-17b-16e-instruct",
            messages=[
                {"role": "system", "content": "You answer only with 'yes'"},
                {"role": "user", "content": prompt}
            ],
            temperature=0
        )
        content = response.choices[0].message.content.strip().lower()
        print("Raw model response:\n", content) # << DEBUG print

        answers = []
        for line in content.splitlines():
            line = line.strip()
            if not line:
                continue
            # Try to extract answer after numbering, fallback if no dot
            if '.' in line:
                ans = line.split('.', 1)[1].strip()
            else:
                ans = line
            if 'yes' in ans:
                answers.append('yes')
            elif 'no' in ans:
                answers.append('no')
            else:
                # Unknown answer, mark 'no' for safety
                answers.append('no')

        if len(answers) == len(texts):
            return answers
        else:
            print(f"Warning: number of answers mismatch (got {len(answers)})")
            return ['no'] * len(texts)
    except:
        pass

```

```

    except openai.error.RateLimitError as e:
        wait_time = (attempt + 1) * 10
        print(f"Rate limit hit. Waiting {wait_time}s before retry...")
        time.sleep(wait_time)
    except Exception as e:
        print(f"OpenAI API error: {e}")
        return ['no'] * len(texts)

    return ['no'] * len(texts)

batch_size = 2 # smaller batch size to reduce prompt token count
df_related = df.copy()

for col in df.columns:
    print(f"Processing column: {col}")
    related_results = []
    col_texts = df[col].fillna("").astype(str).tolist()
    for batch in chunk_list(col_texts, batch_size):
        batch_results = check_batch_openai(batch, col)
        related_results.extend(batch_results)
    df_related[col] = related_results

df_related.to_csv('output_related_check_openai_batched.csv', index=False)
print("Done. Saved to output_related_check_openai_batched.csv")

```

Processing column: Protein URI

Raw model response:

yes

yes

Raw model response:

yes

yes

Raw model response:

yes

yes

Raw model response:

yes

yes

Raw model response:

yes

yes

Processing column: Equivalent URI

Raw model response:

no

no

Raw model response:

no

no

Raw model response:

no

no

Raw model response:

no

no

Raw model response:

no

no

Processing column: owl:annotatedProperty

Raw model response:

here are the answers:

1. no

2. yes

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

to determine if 'owl:annotatedproperty' is related to each text, let's examine them:

1. the first text appears to be an html document header, which does not directly reference or imply 'owl:annotatedproperty'. owl (web ontology language) is used for expressing ontologies, and 'owl:annotatedproperty' is a concept within owl. the provided html snippet does not explicitly mention owl or related concepts.

answer: no

2. pro:dnx seems to refer to a specific identifier or notation, possibly related to biological or chemical nomenclature (pro likely stands for protein research ontology or a similar domain-specific ontology). while this could potentially involve owl or similar ontology languages, there's no direct indication that 'owl:annotatedproperty' is specifically related to "pro:dnx" base

d on the information given.

answer: no

Warning: number of answers mismatch (got 5 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. yes

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

to determine if 'owl:annotatedproperty' is related to each text provided, let's examine them:

1. the first text appears to be an html document header, specifically for a webpage about the protein ontology. it doesn't explicitly mention any owl (web ontology language) terms.

answer: no

2. the second text is an xml header for an rdf (resource description framework) document. it defines several xml namespaces, including one for owl.

although it sets up the namespace for owl ('xmlns:owl="http://www.w3.org/2002/07/owl#"'), it does not directly mention 'owl:annotatedproperty'.

answer: no

Warning: number of answers mismatch (got 6 expected 2), defaulting to 'no'

Processing column: owl:annotatedSource

Raw model response:

to determine if 'owl:annotatedsource' is related to each text provided, let's analyze them:

1. the first text appears to be an html document header, specifically for a webpage about the protein ontology. it doesn't explicitly mention 'owl:annotatedsource', which is a term typically associated with owl (web ontology language) and semantic web annotations.

2. the second text, "reactome:r-hsa-6797641", seems to reference a specific pathway or reaction in the reactome database, which is a resource of biological pathways. the prefix "reactome:" could imply a relation to ontologies or structured biological data, where 'owl:annotatedsource' might be relevant.

given the information:

1. no

2. no

Warning: number of answers mismatch (got 6 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

no.

no.

Raw model response:

here are the answers:

1. no

2. yes

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Processing column: rdfs:comment

Raw model response:

1. no

2. no

Raw model response:

1. yes

2. no

Raw model response:

here are the answers:

1. no

2. no

(rdfs:comment is a property in rdf schema used to provide a human-readable description of a resource. without more context, it seems unlikely that the provided text snippets directly relate to or contain 'rdfs:comment')

Warning: number of answers mismatch (got 4 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

1. no

2. yes

Processing column: oboInOwl:hasBroadSynonym

Raw model response:

to determine if 'oboInOwl:hasBroadSynonym' is related to each text, let's analyze:

1. a protein coding gene acra in dictyostelium discoideum.

- **no**

2. an interleukin-1 receptor type1 that is a translation product of some mrn a whose exon structure and start site selection renders it capable of giving rise to a protein with the amino acid sequence represented by uniprotkb:p135

04-1 or a1:1 ortholog thereof.

- **no**

Warning: number of answers mismatch (got 5 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. yes

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. yes

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

to determine if 'obo:owl:hasbroadsynonym' is related to each text, let's analyze them:

1. **protein ontology home | linked open data | browse | search | statistics | download race-pro pro tracker community pro consortium publications documentation protein ontology report - grb2-related adapter protein2 pr:000008220 - http://purl.obolibrary.org/obo/pr_000008220 protein forms ontology inform**

- the text mentions various ontologies and a specific protein (grb2-related adapter protein2) with an identifier (pr:000008220). the mention of "obo" in the url suggests a connection to the open biological and biomedical ontologies (obo) foundry, which uses the obo-owl (web ontology language) format for its ontologies. 'obo:owl:hasbroadsynonym' is a property that could logically be used within such ontologies to relate terms. **yes**

2. **pro:dnx**

- this seems to refer to a notation or identifier within the protein ontology (pro) or a similar framework. without more context, it's less directly connected to 'obo:owl:hasbroadsynonym' than the first example. however, given that pro is related to protein ontology and likely utilizes obo and owl for its structure, there's still a plausible connection. **yes**

Warning: number of answers mismatch (got 5 expected 2), defaulting to 'no'

Processing column: oboInOwl:hasDbXref

Rate limit hit. Waiting 10s before retry...

Raw model response:

to determine if 'obo:owl:hasdbxref' is related to each text provided, let's analyze them:

1. the text provided appears to be a prefix declaration section of an rdf (resource description framework) or owl (web ontology language) document. it defines various prefixes for well-known vocabularies and standards on the web. 'obo:owl:hasdbxref' does not appear in the snippet provided, but the presence of owl and rdf-related prefixes suggests a context where 'obo:owl:has

dbxref' could potentially be used, as it is part of the obo (open biological and biomedical ontologies) vocabulary often utilized in biomedical ontologies.

****yes****

2. the term 'gene' relates to biological and biomedical contexts, where 'obo:owl:hasdbxref' is often used to provide cross-references to external databases for genes or other biological entities.

****yes****

Warning: number of answers mismatch (got 5 expected 2), defaulting to 'no'
Raw model response:

to determine if 'obo:owl:hasdbxref' is related to each text, let's analyze:

1. the text "asmt; n2bccc89d1394cd0a28ac430d42e7938b1348" seems to represent an identifier or a reference, possibly similar to what 'obo:owl:hasdbxref' would denote, which is a database cross-reference in the context of the obo (open biological and biomedical ontologies) ontology.

2. the text "ne12566163ccc45b3833f59b063a82d9db838" also appears to be an identifier or a reference.

given that 'obo:owl:hasdbxref' pertains to database cross-references:

1. yes

2. yes

Warning: number of answers mismatch (got 6 expected 2), defaulting to 'no'
Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'
Raw model response:

here are the answers:

1. no

2. yes

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'
Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'
Processing column: oboInOwl:hasExactSynonym
Rate limit hit. Waiting 10s before retry...

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'
Raw model response:

1. no

2. no

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

to determine if 'obo:owl:hasExactSynonym' is related to each text, let's analyze them:

1. <http://www.ncbi.nlm.nih.gov/gene/1164905>; fatp-2; sun domain-containing protein4 (arabidopsis thaliana)

– the text provides a gene identifier, a symbol (fatp-2), and a full name (sun domain-containing protein4) specifically for arabidopsis thaliana. the presence of both a symbol and a full name suggests a relationship with ontology terms that might use 'obo:owl:hasExactSynonym' to relate synonyms.

2. mclp1

– this appears to be a simple identifier or acronym for a gene or protein.

given that 'obo:owl:hasExactSynonym' is an owl (web ontology language) property used in biomedical ontologies (like the obo foundry ontologies) to relate a term to its exact synonyms:

1. yes

2. no

Warning: number of answers mismatch (got 8 expected 2), defaulting to 'no'

Raw model response:

to determine if 'obo:owl:hasExactSynonym' is related to each text, let's analyze them:

1. m1 (influenza a virus (strain a/udorn/307/1972 h3n2)) – **no**

2. hgas8/deam:1 – **no**

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Processing column: oboInOwl:hasNarrowSynonym

Rate limit hit. Waiting 10s before retry...

Raw model response:

to determine if 'obo:owl:hasNarrowSynonym' is related to each text, let's analyze:

1. **http://purl.obolibrary.org/obo/pr_000022733; http://purl.obolibrary.org/obo/pr_q9nve5-2; pr:q23490**

– the syntax and structure suggest a relation to obo (open biological and biomedical ontologies) and possibly to specific protein entries.

– obo terms often utilize specific predicates like 'obo:owl:hasNarrowSynonym' for synonym relationships.

2. **http://purl.obolibrary.org/obo/pr_p0acl5; system b(0) neutral amino acid transporter at1 (human)**

– this appears to reference a specific protein (p0acl5) and provides a common name or description ("system b(0) neutral amino acid transporter at1 (human)").

– 'obo:owl:hasNarrowSynonym' could potentially be used to relate such a common name or another synonym to the official identifier.

given the context and typical usage of 'obo:owl:hasnarrowsynonym' in obo and related ontologies for establishing synonym relationships:

1. ****yes****

2. ****yes****

Warning: number of answers mismatch (got 10 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

1. no

2. yes

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Processing column: oboInOwl:hasRelatedSynonym

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. no

Raw model response:

no.

no.

Raw model response:

no.

no.

Raw model response:

to determine if 'obo:owl:hasrelatedsynonym' is related to each text, let's analyze them:

1. the text mentions "protein ontology report – transport and golgi organization protein1 homolog o-glycosylated1 (human) pr:000057376 – <http://purl.obo.library.org/>". this text seems to reference a specific protein ontology entry but does not directly mention 'obo:owl:hasrelatedsynonym'.

****no****

2. the text provides a url: "http://purl.obolibrary.org/obo/pr_000065167;caalm_c112360ca". this url seems to point to an entry in the protein ontology or a similar ontology. the structure suggests it could be related to ontology terms, which might include properties like 'obo:owl:hasrelatedsynonym', b

ut there's no direct indication in the text itself.

****no****

Warning: number of answers mismatch (got 5 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Processing column: rdfs:label

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. yes

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

to determine if 'rdfs:label' is related to each text, let's analyze:

1. http://purl.obolibrary.org/obo/pr_000070511 -

looking at the structure, it seems to follow a pattern typical for identifiers in the obo (open biological and biomedical ontologies) library, which often utilizes rdf (resource description framework) and related vocabularies like rdfs (rdf schema). 'rdfs:label' is commonly used in rdf and owl (web ontology language) ontologies to provide a human-readable label for a resource.

****yes****

2. http://purl.obolibrary.org/obo/pr_q8k0d0 -

similarly, this identifier follows the same pattern and context as the first one, suggesting it is also part of the obo library. given the context and common practices in ontology development, 'rdfs:label' would typically be used here as well.

****yes****

Warning: number of answers mismatch (got 7 expected 2), defaulting to 'no'

Raw model response:

1. no

2. no

Processing column: owl:onProperty

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. no

2. no

Raw model response:

1. yes

2. no

Processing column: owl:someValuesFrom

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. no

Raw model response:

1. no

2. no

Raw model response:

1. no

2. no

Raw model response:

to determine if 'owl:somevaluesfrom' is related to each text, let's briefly consider what 'owl:somevaluesfrom' means. it is a construct in owl (web ontology language) used to describe a property restriction. specifically, it denotes that for a given property, there exists at least one value that the property relates to.

given the context:

1. ****m1 (influenza a virus (strain a/udorn/307/1972 h3n2)); pro:dnx**** – this line seems to provide a specific instance (m1) of an influenza a virus strain with an accession or identifier (pro:dnx). without more context about how 'owl:somevaluesfrom' is being used in relation to this text, we can't directly infer a connection based solely on the information given. however, if this text is part of a larger ontology or description where properties of the influenza a virus are being defined (e.g., "has strain"), then 'owl:somevaluesfrom' could potentially be used to express that a certain property (like "has strain") has at least one value (in this case, "a/udorn/307/1972 h3n2").

****answer: yes****

2. ****aqr (mouse); nf434fd27b84b4345befbd5526f317f54b1061**** – similar to the first example, without explicit context on how 'owl:somevaluesfrom' relates to this text, we can't directly confirm a connection. however, if aqr (mouse) is being described with properties (e.g., "has identifier"), then 'owl:somevaluesfrom' could be relevant if stating that aqr (mouse) has at least one identifier.

****answer: yes****

Warning: number of answers mismatch (got 6 expected 2), defaulting to 'no'

Raw model response:

1. yes

2. no

Processing column: rdfs:subClassOf
Rate limit hit. Waiting 10s before retry...
Raw model response:
here are the answers:

1. no
2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

1. no
2. no

Raw model response:

1. no
2. no

Raw model response:

to determine if 'rdfs:subclassof' is related to each text, let's analyze them in the context of what 'rdfs:subclassof' means. 'rdfs:subclassof' is a relationship in rdf schema (rdfs) that indicates one class is a subclass of another. it implies that every instance of the first class is also an instance of the second class.

1. ****a protein that is a translation product of the human osgepl1 gene or a 1:1 ortholog thereof.****

- this statement describes a specific type of protein related to the osgepl1 gene or its ortholog. in a semantic or ontological context, this could be represented as a class or a type of protein. if we consider "protein" as a class, then "a protein that is a translation product of the human osgepl1 gene or a 1:1 ortholog thereof" could be considered a subclass of "protein." therefore, 'rdfs:subclassof' could relate to this text if it's describing class hierarchies.

- ****answer: yes****

2. ****a protein coding gene ipo8 in zebrafish.; n4c641700d6bb46049e644d5ff510cf6cb3402****

- this statement seems to describe a specific gene (ipo8) in zebrafish, identified by a unique string (n4c641700d6bb46049e644d5ff510cf6cb3402). while this does refer to a biological entity (a gene), it does not explicitly describe a class relationship like 'rdfs:subclassof' implies. it seems more like an instance or an individual rather than a class definition.

- ****answer: no****

Warning: number of answers mismatch (got 7 expected 2), defaulting to 'no'

Raw model response:

to determine if 'rdfs:subclassof' is related to each text, let's consider what 'rdfs:subclassof' means. it is a relationship in rdf schema (rdfs) that indicates one class is a subclass of another. this implies a specific ontological relationship often used in knowledge graphs and semantic web applications, particularly in biological ontologies like the gene ontology.

1. ****sun domain-containing protein 4 (arabidopsis thaliana)**** - this appears to be a specific protein from a specific organism (arabidopsis thaliana). the term suggests a classification or identification of a protein based on its domain (sun domain) and its source organism.

- ****yes** or **no**: *no***

2. `**a gamma-adducin that is encoded in the genome of rat.**` – this describes a specific protein (gamma-adducin) from a specific organism (rat), identified by its function or classification and genetic origin.

– `**yes**` or `**no**`: `*no*`

in both cases, `'rdfs:subclassof'` would relate if the descriptions directly pertain to class-subclass relationships within an ontology. however, the provided texts seem more like specific instance descriptions rather than class definitions. therefore, based on the information given and typical usage, the direct relationship to `'rdfs:subclassof'` seems unlikely for these specific examples as they seem to represent instances or types rather than explicitly stating class-subclass relationships.

however, re-evaluating strictly:

- the first can be seen as an instance or a product of a class (protein) but does not directly state a class-subclass relationship.
- the second similarly describes an instance.

given standard interpretations in ontology contexts:

1. `*no*`

2. `*no*`

Warning: number of answers mismatch (got 12 expected 2), defaulting to 'no'

Processing column: `rdf:type`

Rate limit hit. Waiting 10s before retry...

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

here are the answers:

1. no

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

to determine if `'rdf:type'` is related to each text, let's analyze:

1. the text mentions `'http://www.w3.org/2002/07/owl#class'`, which implies a classification or type definition in owl (web ontology language). `'rdf:type'` is used in rdf (resource description framework) to state that a resource is of a certain type. owl builds upon rdf, and when defining classes in owl, `'rdf:type'` can be used to relate instances to their classes.

2. similarly, the second text also mentions `'http://www.w3.org/2002/07/owl#class'`, indicating the same context of class definition in owl.

given that `'rdf:type'` is a predicate used to relate a resource to its type, and both texts reference owl classes, `'rdf:type'` can be considered related to

o both as it can be used to assert the type of a resource in the context of owl ontologies.

1. yes

2. yes

Warning: number of answers mismatch (got 6 expected 2), defaulting to 'no'

Raw model response:

to determine if 'rdf:type' is related to each text, let's analyze:

1. the urls provided are from specific databases (pro consortium and dictybase) and reference specific entries within those databases. the mention of obo (open biological and biomedical ontologies) terms (e.g., 'http://purl.obolibrary.org/obo/pr#pro-short-label' and 'http://purl.obolibrary.org/obo/pr#000044509') indicates a connection to semantic web technologies and ontologies, where rdf (resource description framework) and its types ('rdf:type') are fundamental.

given this context, ****yes****.

2. similarly, the second set of urls also pertains to biological databases and obo terms, suggesting a similar connection to semantic web frameworks and ontologies.

given this context, ****yes****.

Warning: number of answers mismatch (got 5 expected 2), defaulting to 'no'

Processing column: Equivalent URI_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. no

2. yes

Raw model response:

1. no

2. yes

Raw model response:

1. no

2. no

Processing column: owl:annotatedProperty_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. no

Processing column: owl:annotatedSource_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. no

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. no

Processing column: rdfs:comment_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. no

Raw model response:

1. yes

2. no

Processing column: oboInOwl:hasBroadSynonym_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. yes

2. yes

Raw model response:

1. no

2. no

Raw model response:

1. no

2. no

Raw model response:

1. no

2. no

Raw model response:

1. no

2. yes

Processing column: oboInOwl:hasDbXref_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. yes

Raw model response:

1. yes

2. yes

Raw model response:

here are the answers:

1. yes

2. no

Warning: number of answers mismatch (got 3 expected 2), defaulting to 'no'

Raw model response:

1. no

2. no

Processing column: oboInOwl:hasExactSynonym_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. yes

Raw model response:

1. yes

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. yes

2. yes

Processing column: oboInOwl:hasNarrowSynonym_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. no

2. no

Raw model response:

1. no

2. yes

Raw model response:

1. no

2. no

Raw model response:

1. no

2. no

Raw model response:

1. yes

2. no

Processing column: oboInOwl:hasRelatedSynonym_uri_valid

Rate limit hit. Waiting 10s before retry...

Raw model response:

1. yes

2. yes
Raw model response:
1. no
2. yes
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Processing column: rdfs:label_uri_valid
Rate limit hit. Waiting 10s before retry...
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Raw model response:
1. yes
2. yes
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Processing column: owl:onProperty_uri_valid
Rate limit hit. Waiting 10s before retry...
Raw model response:
1. no
2. yes
Raw model response:
1. no
2. yes
Raw model response:
1. no
2. yes
Raw model response:
1. no
2. yes
Processing column: owl:someValuesFrom_uri_valid
Rate limit hit. Waiting 10s before retry...
Raw model response:
1. yes
2. yes
Raw model response:
1. yes
2. yes
Raw model response:

1. yes
2. yes
Raw model response:
1. yes
2. yes
Raw model response:
1. no
2. yes
Processing column: rdfs:subClassOf_uri_valid
Rate limit hit. Waiting 10s before retry...
Raw model response:
1. yes
2. yes
Raw model response:
1. yes
2. yes
Raw model response:
1. yes
2. no
Raw model response:
1. yes
2. yes
Raw model response:
1. yes
2. yes
Processing column: rdf:type_uri_valid
Rate limit hit. Waiting 10s before retry...
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Raw model response:
1. no
2. no
Done. Saved to output_related_check_openai_batched.csv